

An open-source unconstrained stress updating algorithm for the modified Cam-clay model

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Abstract

This paper presents an unconstrained stress updating algorithm for a critical state plastic model of clay soil, where the loading/unloading estimations and the consideration of the stress behaviour transition from elasticity to plasticity can be bypassed by using the *Fischer–Burmeister* smoothing function instead of the *Kuhn–Tucker* complementarity conditions. A smoothing tangent operator consistent with the unconstrained stress updating strategy is derived from preserving the quadratic convergence speed for the global solution. Specifically, the relationship and difference between the consistent and continuum tangent operators are analysed from the perspectives of algebra and geometry. In addition, the nonlinear constitutive equations obtained by the backward Euler integration scheme are solved by the double dogleg trust region method (improved by non-monotonic technology), where a larger strain increment than that of the newton method is allowed for the stress updating. Then, the modified Cam-clay model for soil is implemented in ABAQUS/Standard by the proposed algorithm. The results of numerical examples show that the algorithm has significant advantages in terms of computation efficiency and robustness in contrast to the ABAQUS/Standard default integration algorithm, especially for the condition of large load increments and cyclic loadings. More than twice the computational efficiency of the ABAQUS/Standard default integration algorithm can be observed in the representative numerical examples. The source code of the proposed algorithm is freely available at https://github.com/zhouxin615/NMTR_Method.

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1. Introduction

As the landmark product for the development of critical state soil mechanics [1], the modified Cam clay (MCC) model is one of the origins of numerous elastoplastic models for soil [2–5] and plays an essential role in the numerical analysis of geotechnical engineering problems [6]. Accordingly, the stress updating algorithm serving used for numerical implementations of the MCC model has always been considered [7–9], as it provides a crucial tool for local calculations in finite element analysis.

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Due to the presence of the plastic deformation mechanism of soil, the MCC model is presented in the form of rate constitutive equations that are constrained by *Kuhn–Tucker* (*KT*) complementary conditions, where the deformation behaviour of the material is constrained by the loading/unloading inequality. During the stress updating process, the rate constitutive equations are first discretized into nonlinear algebraic equations in time by certain integration schemes, e.g., the forward Euler approach, modified Euler approach, Runge–Kutta technique [9], and Next Increment Corrects Error method following the explicit scheme [10], as well as the backward Euler method and trapezoidal and midpoint rules following the implicit scheme [11]. Then, the updated stress can be obtained at the material point after the following two crucial issues are addressed: (i) How can the updating process of state variables satisfy the constraints of the loading/unloading inequality? (ii) How can nonlinear constitutive equations be solved efficiently and robustly when the implicit integration scheme is used?

The most commonly used method for addressing loading/unloading inequality constraints is the operator splitting technique [11]. Namely, the elastic prediction is first completed by freezing the plastic flow, and then whether to accept the trial stress as the new stress or to carry out a plastic correction depends on whether the trial stress violates the *KT* conditions. Operator splitting technology provides a standard computational paradigm for the elastoplastic model, but as necessary efforts, i.e., loading/unloading judgments based on the trial stress, need to be done, additional consideration is required for the case in which the stress point undergoes a transition from elastic to plastic behaviour when the explicit integration scheme is used. There are also some alternatives to this computational framework. In Zheng et al.'s recent pioneering work [12], the control equations of the Mohr–Coulomb model with perfect plasticity were cast as a certain variational inequality that is equivalent to the mixed complementarity problem and was then solved effectively by the projection and contraction method [13]. However, the ability of this projection and contraction method [13] in dealing with the softening problem remains to be explored. Another more common idea is to transform inequality constraints into equality constraints using specific functions, such as the penalty function [8,14,15] and the smoothing function [16–18]. The former is typically used in conjunction with the interior point method in analysis dealing with elastoplastic problems from the perspective of mathematical programming [19], where the constitutive model is recast in terms of a variational principle, and then the inequality constraints are transformed into equality constraints by introducing the appropriate penalty functions. The details of applying this method to the MCC model can be found in the literature [8]. The latter uses a smoothing function to replace non-smoothing *KT* conditions, which makes the elastic and plastic behaviours uniformly described by a set of smooth nonlinear equations. In contrast to that of the penalty function, the advantage of the smoothing function is that it does not increase the number of required nonlinear equations. Notably, there is also no need for additional computational effort to address the loading/unloading judgment and stress behaviour transition processes in the operator splitting technique. Smoothing functions have been successfully used in the analysis of both crystal plasticity [16,20] and finite strain plasticity [17,18,21] due to their attractive simplicity and effectiveness. There is no doubt that such a function also provides a concise and valid computational framework for the implementation of the MCC model based on plastic flow theory. It is also a prospective research direction for solving the stress updating problem faced by the plastic models of geotechnical materials.

For obtaining the solutions of nonlinear equations, the newton method is one of the most commonly used mathematical tools due to its quadratic convergence rate. It has been widely used in some representative implicit stress updating algorithms, such as the fully implicit return-mapping algorithm [22], the cutting-plane algorithm [23,24], and the semi-implicit algorithm [25]. However, the convergence behaviour of the newton method relies crucially on the proximity of the initial iteration point to the convergence point [26]. The pure newton method is not guaranteed to converge for a large time step or complex models with strong nonlinearity, although this drawback can be compensated to a certain extent by breaking a large input load into a series of smaller load increments and sequentially solving the sub-loadings [27]. From the perspective of mathematical programming, the nonlinear equation problem is equivalent to an unconstrained minimization problem [28], which can be solved effectively by some mature optimization methods, e.g., the line search method and the trust region method [29,30]. In some sense, the search strategies for the two methods are dual to each other. The former optimizes the step size with the given search direction, while the latter first choose the step size (i.e., trust region radius) and then determines the search direction. The two methods have been utilized to attempt to solve the stress updating issue of elastoplastic problems due to their better convergence than that of the classical Newton method. Some very valuable studies in this field can be found in the literature [30–35]. Compared with the line search method, however, the application of trust region method in the stress updating calculation is rarely explored. To the authors' knowledge, it seems

that no one studied the performance of this method in the elastoplastic soil models. Especially, Nocedal and Wright also pointed out that trust region algorithms may have certain theoretical advances in solving nonlinear equations problem compared with the line search method in their monograph on the numerical optimization methods [28]. Although the trust region method has shown excellent convergence and robustness in solving the problem of the nonlinear equation, the tentative application of the trust region method to elastoplastic models still encounters some expected computation difficulties in some cases [30,33] (e.g., high curvature and poor scaling problems) due to the limitation existing in the original version of trust region method. These computational difficulties can also be solved by introducing some improvement techniques from the optimization field [30,36–38] to obtain a more satisfactory algorithmic stress updating performance for elastoplastic problems.

After the stress in the local calculation is updated, the consistent tangent operator (CTO) from the material point is required to form the global stiffness of the structure, which is capable of preserving the asymptotic quadratic convergence speed of the global solution. The origin of the concept of the CTO can be traced back to work by Hughes [39]. It was then discussed in the analysis of inelastic large strain problems by Nagtegaal [40] and was developed in the small strain framework by Simo [41]. In some follow-up studies, the CTO has always been an important component of stress updating algorithms [23,27,42–44]. Specifically, for the MCC model, the CTO in conjunction with the implicit return-mapping stress updating algorithm was derived in a study by Borja and his co-workers [7,45] and appeared to have excellent numerical performance. However, one must realize that within the computational framework of the operator splitting technique, the CTO needs to be derived respectively by elastic and elastoplastic constitutive equations to address loading and unloading problems. Under repeated loading and unloading conditions, the convergence of global solution may also be difficult to the optimal state. The reason for this is that the non-smoothness of the elastoplastic problem resulting from the loading/unloading constraint inequalities is not truly eliminated by the operator splitting technique, and it may impede the convergence of the global solution under repeated loading and unloading conditions. However, this limitation can also be addressed by using a smoothing function instead of non-smooth KT conditions.

In this paper, an unconstrained stress updating strategy without the need for loading/unloading judgments is proposed. The nonlinear constitutive equations are solved by the non-monotonic trust region (NMTR) method with strong convergence and robustness. The poor scaling problem encountered by the trust region method according to the literature [33] is eliminated by introducing a diagonal scaling matrix without requiring additional parameters [36,37]. The smoothing CTO corresponding to the unconstrained stress updating strategy is derived to preserve the quadratic convergence speed of the global solution when using the newton method. After applying the proposed algorithm to the MCC model, a numerical validation shows that its performance is superior to that of the default algorithm (ABAQUS/Standard). The remainder of the paper is organized as follows: Section 2 focuses on the elaboration of the basic idea of an unconstrained stress updating strategy with a simple 1D elastoplastic physical model, in which the operator splitting technique is briefly reviewed as a comparison. Section 3 briefly restates the MCC model and provides the integration formulas derived by the backward Euler scheme. The double dogleg trust region method is improved by the non-monotonic technique and presented in Section 4. In Section 5, the algebraic and geometric interpretations of the CTO and continuum tangent operator (CON) are discussed in depth, and the two tangent operators that are consistent with the unconstrained stress updating strategy are derived. Section 6 gives the necessary numerical validation and performance analysis for the proposed algorithm based on the user-defined material subroutine (UMAT). Some important conclusions are contained in Section 7. In the [Appendices](#), we give some derivation details and the matrix representation of tensors, which is convenient for programming. Finally, we add a simple clarification to the following contents. The constitutive models targeted by the algorithm are in a small strain theory framework. This paper takes pressure as positive. Tensors are represented in bold. Matrices and vectors are denoted by $[\cdot]$ and $\{\cdot\}$, respectively. The three indices n , i , and k are used to denote the incremental load step, the global equilibrium iteration, and the local stress updating iteration, respectively.

2. Computational framework of the unconstrained stress updating strategy

To completely describe the material's mechanical response, a basic elastoplastic model is required to answer the following questions: What is the stress–strain relationship of the material? How to determine the proportional relationship between plastic strain increment components? How to record the loading history related to plastic

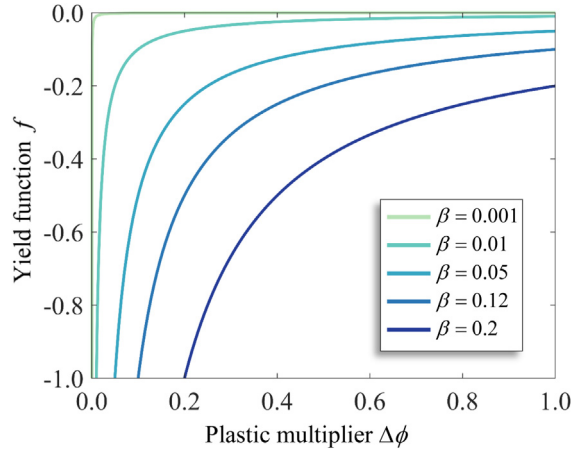


Fig. 1. Influence law of the parameter β on the FB smoothing function (with $c_d = 1$).

deformation? What situations will trigger the plastic deformation mechanism? The answers to these questions constitute the basic mathematical formula of the elastoplastic model:

$$\begin{cases} \dot{\boldsymbol{\sigma}} = \mathbf{D} : \dot{\boldsymbol{\epsilon}}^e = \mathbf{D} : (\dot{\boldsymbol{\epsilon}} - \dot{\boldsymbol{\epsilon}}^p), & \text{Hooke's law} \\ \dot{\boldsymbol{\epsilon}}^p = \dot{\phi} \mathbf{r}, & \text{Plastic flow rule} \\ \dot{\alpha} = \dot{\phi} h, & \text{Hardening law} \\ \dot{\phi} \geq 0, f(\boldsymbol{\sigma}, \alpha) \leq 0, \dot{\phi} f(\boldsymbol{\sigma}, \alpha) = 0, & \text{Kuhn-Tucker complementary conditions} \end{cases} \quad (1)$$

where $\dot{\boldsymbol{\sigma}}$ and $\dot{\boldsymbol{\epsilon}}$ denote the rates of stress tensor and strain tensor, respectively. The latter includes recoverable elastic part $\dot{\boldsymbol{\epsilon}}^e$ and unrecoverable plastic part $\dot{\boldsymbol{\epsilon}}^p$. $\mathbf{D}$ is the elastic stiffness tensor. $\dot{\phi}$ and $\mathbf{r}$ are the rate of the plastic multiplier and the plastic flow direction, respectively. $\dot{\alpha}$ and h are the rate of plastic internal variable and its gradient. f denotes the yield function defining the elastic and plastic boundary. In the numerical implementation, these rate equations are transformed into the algebraic equations by time integral:

$$\begin{cases} \boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}_n + \mathbf{D} : (\Delta \boldsymbol{\epsilon}_{n+1} - \Delta \boldsymbol{\epsilon}_{n+1}^p) \\ \Delta \boldsymbol{\epsilon}_{n+1}^p = \Delta \phi_{n+1} \mathbf{r}_{n+1} \\ \Delta \alpha_{n+1} = \Delta \phi_{n+1} h_{n+1} \\ \Delta \phi_{n+1} \geq 0, f_{n+1} \leq 0, \Delta \phi_{n+1} f_{n+1} = 0 \end{cases} \quad (2)$$

where the backward Euler integration scheme is used as an example. As can be noticed, the above equations are non-smooth due to the constraints of inequalities. Alternatively, the *KT* conditions in Eq. (2) can be replaced equivalently by the smooth function in the optimization theory:

$$\begin{cases} \boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}_n + \mathbf{D} : (\Delta \boldsymbol{\epsilon}_{n+1} - \Delta \boldsymbol{\epsilon}_{n+1}^p) \\ \Delta \boldsymbol{\epsilon}_{n+1}^p = \Delta \phi_{n+1} \mathbf{r}_{n+1} \\ \Delta \alpha_{n+1} = \Delta \phi_{n+1} h_{n+1} \\ \sqrt{(c_d \Delta \phi_{n+1})^2 + f_{n+1}^2} + 2\beta - c_d \Delta \phi_{n+1} + f_{n+1} = 0 \end{cases} \quad (3)$$

where Eq. (3)₄ is the *Fischer–Burmeister (FB)* smooth function [46,47]. β is the parameter controlling the approximation degree of Eq. (3)₄ to Eq. (2)₄, as shown in Fig. 1. The parameter c_d is used to balance the dimensional relationship between $\Delta \phi_{n+1}$ and f_{n+1} . Then, the non-smoothness of the elastoplastic problem is eliminated in Eq. (3). The original constrained problem is transformed into the equivalent unconstrained problem.

In what follows, a 1D spring-slider physical model shown in Fig. 2 is used as an example to further illustrate the computational framework of the unconstrained stress updating strategy. The classical splitting stress updating

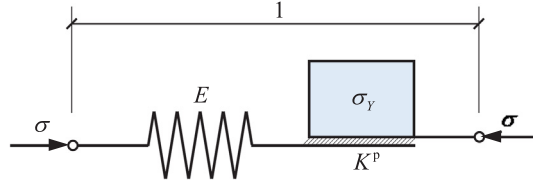


Fig. 2. 1D spring-slider physical model.

strategy is also briefly reviewed for comparison. Corresponding to Eq. (2), the integral equations of the 1D physical model are given:

$$\begin{cases} \sigma_{n+1} = \sigma_n + E (\Delta \varepsilon_{n+1} - \Delta \varepsilon_{n+1}^p) \\ \Delta \varepsilon_{n+1}^p = \Delta \phi_{n+1} \text{sign}(\sigma_{n+1}) \\ \Delta \alpha_{n+1} = |\Delta \varepsilon_{n+1}^p| \\ \Delta \phi_{n+1} \geq 0, f(\sigma_{n+1}, \alpha_{n+1}) \leq 0, \text{ and } \Delta \phi_{n+1} f(\sigma_{n+1}, \alpha_{n+1}) = 0 \end{cases} \quad (4)$$

where $f(\sigma, \alpha) = |\sigma| - \sigma_Y - K^P \alpha$ and the simple linear hardening law is assumed. E and K^P are the spring stiffness and the hardening modulus of the slider. σ_Y is the initial sliding stress. $\text{sign}(\cdot)$ is the sign function and Eq. (4)₂ denotes that the direction of $\Delta \varepsilon_{n+1}^p$ is the same as that of stress in the 1D condition. When only the compression situation is considered, the sign and absolute value functions can be omitted.

The key to solving the elastoplastic problem defined by Eq. (4) is to determine $\Delta \phi_{n+1}$ under the action of an external load. However, for the given $\Delta \varepsilon_{n+1}$, three loading/unloading cases may need to be estimated before calculating $\Delta \phi_{n+1}$ due to the presence of KT conditions, as demonstrated in Fig. 3.

2.1. Operator splitting stress updating strategy

In the classical return-mapping stress updating algorithm, the operator splitting technique [11] is used to address KT conditions. First, the trial stress $\sigma_{n+1}^{trial} = \sigma_n + E \Delta \varepsilon_{n+1}$ is determined based on the elastic Hooke's Law. Then, the loading and unloading cases under the current increment step can be estimated by the values of $f(\sigma_n, \alpha_n)$ and $f(\sigma_{n+1}^{trial}, \alpha_n)$ as follows:

$$\begin{cases} \text{case } a: \text{pure elastic loading/unloading} & \text{if } f(\sigma_n, \alpha_n) \leq 0 \text{ and } f(\sigma_{n+1}^{trial}, \alpha_n) \leq 0 \\ \text{case } b: \text{pure elastoplastic loading} & \text{if } f(\sigma_n, \alpha_n) = 0 \text{ and } f(\sigma_{n+1}^{trial}, \alpha_n) > 0 \\ \text{case } c: \text{elastic and elastoplastic mixed loading} & \text{if } f(\sigma_n, \alpha_n) < 0 \text{ and } f(\sigma_{n+1}^{trial}, \alpha_n) > 0 \end{cases} \quad (5)$$

For case a , that the current increment step is purely elastic. $\Delta \phi_{n+1} = 0$ and the σ_{n+1}^{trial} is accepted as the σ_{n+1} . For case b , the plastic deformation mechanism is triggered under the current increment step. The plastic corrector needs to be used to pull the trial stress back toward the real stress σ_{n+1} , as shown in Fig. 4. $\Delta \phi_{n+1}$ can be obtained by substituting Eq. (4)_{1~3} and $\sigma_n = \sigma_Y + K^P \alpha_n$ into the equation $f(\sigma_{n+1}, \alpha_{n+1}) = \sigma_{n+1} - \sigma_Y - K^P \alpha_{n+1}$ as follows:

$$\Delta \phi_{n+1} = \frac{E \Delta \varepsilon_{n+1}}{E + K^P} \quad (6)$$

Case c indicates that the stress point undergoes a mechanical behaviour transition from elasticity to plasticity at the current increment step. The partial strain increment $\xi \Delta \varepsilon_{n+1}$ corresponding to the pure elastic loading in Fig. 3(c) needs to be determined in advance. The scale factor ξ ($0 < \xi < 1$) is determined by solving the equation $f(\sigma_{int}, \alpha_n) = 0$ and considering the condition $\sigma_n < \sigma_Y + K^P \alpha_n$, where σ_{int} is the largest historical yield stress:

$$\xi = \frac{\sigma_Y + K^P \alpha_n - \sigma_n}{E \Delta \varepsilon_{n+1}} \quad (7)$$

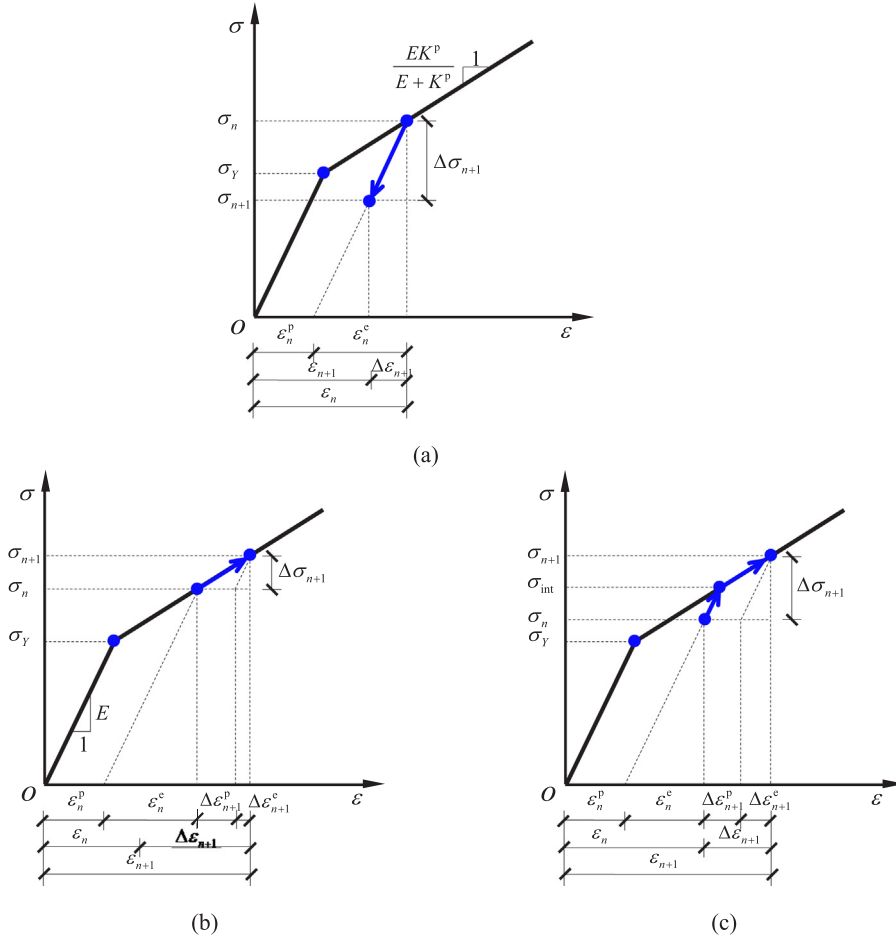


Fig. 3. Different loading/unloading cases: (a) pure elastic loading/unloading; (b) pure elastoplastic loading; (c) mixed elastic and elastoplastic loading.

Then, $\Delta\phi_{n+1}$ corresponding to case c can be obtained by replacing $\Delta\epsilon_{n+1}$ in Eq. (6) with $(1 - \xi) \Delta\epsilon_{n+1}$:

$$\Delta\phi_{n+1} = \frac{E \Delta\epsilon_{n+1} - (\sigma_Y + \alpha_n K^p - \sigma_n)}{E + K^p} \quad (8)$$

Finally, the updating of state variables for the different loading and unloading cases can be obtained by substituting $\Delta\phi_{n+1} = 0$, Eqs. (6) and (8) into Eq. (4) as follows:

For case a:

$$\begin{cases} \sigma_{n+1} = \sigma_n + E \Delta\epsilon_{n+1} \\ \alpha_{n+1} = \alpha_n \\ \Delta\phi_{n+1} = 0 \end{cases} \quad (9)$$

For case b:

$$\begin{cases} \sigma_{n+1} = \sigma_n + \frac{E K^p \Delta\epsilon_{n+1}}{E + K^p} \\ \alpha_{n+1} = \alpha_n + \frac{E \Delta\epsilon_{n+1}}{E + K^p} \\ \Delta\phi_{n+1} = \frac{E \Delta\epsilon_{n+1}}{E + K^p} \end{cases} \quad (10)$$

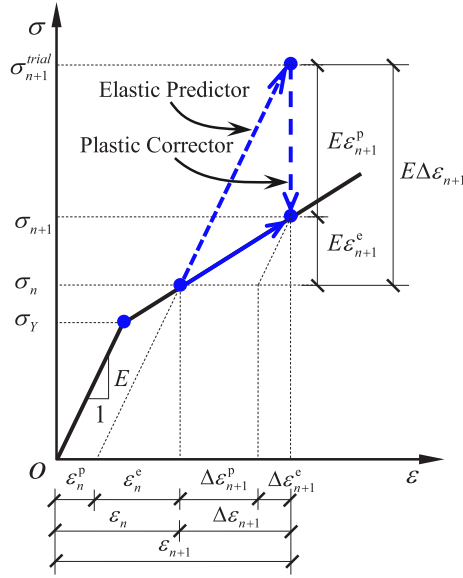


Fig. 4. Stress update procedure for the purely elastoplastic loading case.

For case c:

$$\begin{cases} \sigma_{n+1} = \sigma_n + \frac{EK^p \Delta \varepsilon_{n+1}}{E + K^p} + \frac{E(\sigma_Y + \alpha_n K^p - \sigma_n)}{E + K^p} \\ \alpha_{n+1} = \alpha_n + \frac{E \Delta \varepsilon_{n+1} - (\sigma_Y + \alpha_n K^p - \sigma_n)}{E + K^p} \\ \Delta \phi_{n+1} = \frac{E \Delta \varepsilon_{n+1} - (\sigma_Y + \alpha_n K^p - \sigma_n)}{E + K^p} \end{cases} \quad (11)$$

In the operator splitting stress updating strategy, the loading/unloading judgment is achieved by the elastic predictor. Although this method provides an excellent distinction between the different loading/unloading cases, it leads to non-smoothness in the calculation. Additional attention needs to be paid to case c (if the explicit integration scheme is adopted). It can be found that the root of the non-smoothness in the KT conditions arises from the inequality constraints. Replacing the inequality constraints with a smoothing function to eliminate the non-smoothness in the calculation becomes the biggest motivation of this paper.

2.2. Unconstrained stress updating strategy based on a smoothing function

Replacing Eq. (4)₄ with the FB smooth function, the elastoplastic problem constrained by the loading/unloading inequalities can be transformed into the following unconstrained form:

$$\begin{cases} \sigma_{n+1} = \sigma_n + E(\Delta \varepsilon_{n+1} - \Delta \varepsilon_{n+1}^p) \\ \Delta \varepsilon_{n+1}^p = \Delta \phi_{n+1} \\ \Delta \alpha_{n+1} = \Delta \varepsilon_{n+1}^p \\ \sqrt{(c_d \Delta \phi)^2 + f^2(\sigma_{n+1}, \alpha_{n+1})} + 2\beta - c_d \Delta \phi_{n+1} + f(\sigma_{n+1}, \alpha_{n+1}) = 0 \end{cases} \quad (12)$$

Now, the elastoplastic problem depicted by Fig. 2 can be solved directly by Eq. (12) without dealing with inequality constraints. For Eq. (12)₄, we have:

$$\beta = -c_d \Delta \phi_{n+1} f(\sigma_{n+1}, \alpha_{n+1}) \quad (13)$$

Substituting the condition $f(\sigma_{n+1}, \alpha_{n+1}) = \sigma_n + E(\Delta\varepsilon_{n+1} - \Delta\phi_{n+1}) - \sigma_Y - K^p(\alpha_n + \Delta\phi_{n+1})$ into Eq. (13), we can obtain:

$$c_d \Delta\phi_{n+1}^2 (E + K^p) - c_d \Delta\phi_{n+1} (\sigma_n + E\Delta\varepsilon_{n+1} - \sigma_Y - \alpha_n K^p) - \beta = 0 \quad (14)$$

Utilizing the root of the quadratic equation, the nonnegative solutions of $\Delta\phi_{n+1}$ can be obtained as follows:

$$\Delta\phi_{n+1} = \frac{c_d (\sigma_n + E\Delta\varepsilon_{n+1} - \sigma_Y - \alpha_n K^p) + \sqrt{c_d^2 (\sigma_n + E\Delta\varepsilon_{n+1} - \sigma_Y - \alpha_n K^p)^2 + 4c_d (E + K^p) \beta}}{2c_d (E + K^p)} \quad (15)$$

When β is close to 0, the following solutions for $\Delta\phi_{n+1}$ can be automatically obtained for a given $\Delta\varepsilon_{n+1}$ and σ_n :

$$\lim_{\beta \rightarrow 0} \Delta\phi_{n+1} = \begin{cases} 0, & \sigma_n \leq \sigma_Y + \alpha_n K^p \text{ and } \sigma_n + E\Delta\varepsilon_{n+1} \leq \sigma_Y + \alpha_n K^p \\ \frac{E\Delta\varepsilon_{n+1}}{E + K^p}, & \sigma_n = \sigma_Y + \alpha_n K^p \text{ and } \sigma_n + E\Delta\varepsilon_{n+1} > \sigma_Y + \alpha_n K^p + E\Delta\varepsilon_{n+1} \\ \frac{E\Delta\varepsilon_{n+1} - (\sigma_Y + \alpha_n K^p - \sigma_n)}{E + K^p}, & \sigma_n < \sigma_Y + \alpha_n K^p \text{ and } \sigma_n + E\Delta\varepsilon_{n+1} > \sigma_Y + \alpha_n K^p + E\Delta\varepsilon_{n+1} \end{cases} \quad (16)$$

where there is no need to perform the loading/unloading judgment in advance or to calculate the intersection point σ_{int} in Fig. 3(c). Substituting Eq. (16) into Eq. (12), the updating formulas of the same state variable as Eqs. (9), (10), and (11) can be obtained correctly for the given $\Delta\varepsilon_{n+1}$ even if the operator splitting technique is not used. The non-smoothness of the elastoplastic problem is bypassed by the *FB* smoothing function.

It is noted that the constitutive equations for the elastoplastic problems in this section are solved manually due to the simplicity of the 1D physical model. However, the mechanical behaviours of most materials, especially for geomaterials, are often complex and exhibit highly nonlinear characteristics. The nonlinear stress integrations need to be solved by iteration. In addition, the smoothing function also results in a problem that we must face, that is, the smoothing function may have a higher curvature near the origin than elsewhere. In turn, the high curvature phenomenon appears in the area near $f_{n+1} = 0$ and $\Delta\phi_{n+1} = 0$ when β trends to 0. This makes it more difficult to address the problem of neutral loading and elastic unloading, where the initial stress point appears on the yield surface. Therefore, an efficient and robust numerical algorithm is required to solve the nonlinear constitutive equations of the complex elastoplastic model containing the smoothing function.

3. Backward Euler integral scheme of the MCC model based on the smoothing function

The initial Cam-clay model was proposed by Roscoe and his co-workers in the 1960s [48], and it has directly affected the beginning and development of critical state soil mechanics and became a standard paradigm for the successor to establish the elastoplastic model of soils [49,50]. In this section, the MCC model developed further by Roscoe and Burland [51] is considered as the application object of the proposed algorithm. The rate constitutive equations are integrated by the backward Euler scheme within the framework of an unconstrained stress updating strategy.

3.1. Kernel ingredients of the model

The basic architecture of the MCC model can be represented by four key ingredients, including the nonlinear elastic law for soil, the elliptic yield function, the associated flow rule, and the hardening law.

3.1.1. Elastic law

Under isotropic conditions, the fourth-order elastic stiffness tensor in Eq. (1) is expressed as follows:

$$\mathbf{D} = 3\mathbf{I}^{\text{vol}} \left(K - \frac{2}{3}G \right) + 2\mathbf{I}^{\text{sym}}G \quad (17)$$

where $\mathbf{I}^{\text{vol}}$ and $\mathbf{I}^{\text{sym}}$ are the volume part and symmetric part of the fourth-order unit tensor $\mathbf{I}$, respectively. G and K denote the shear modulus and the bulk modulus, respectively. $G = Kr$, and $r = 1.5(1 - 2\nu)/(1 + \nu)$ where ν is

Poisson's ratio. In the MCC model, the following expression is employed to describe the nonlinear elastic behaviour of the soil:

$$K = c_k p \quad (18)$$

where $c_k = (1 + e_0) / \kappa$. e_0 is the initial void ratio, and κ is the swell index under isotropic compression. $p = \boldsymbol{\sigma} : \mathbf{1} / 3$ denotes the hydrostatic pressure where $\mathbf{1}$ denotes the second-order unit tensor.

3.1.2. Yield function

The following elliptic yield function is used in the MCC model:

$$f = \frac{q^2}{M^2} + p(p - p_c) \quad (19)$$

where M is the slope of the critical state line. $q = \sqrt{3/2} \|\mathbf{s}\|$ denotes the generalized shear stress, and $\mathbf{s} = \boldsymbol{\sigma} - p\mathbf{1}$ is the deviatoric stress tensor. p_c denotes the pre-consolidation pressure, which defines the diameter along the hydrostatic axis of the ellipsoid yield surface.

3.1.3. Flow rule

For the MCC model, the associated flow rule, which assumes that the plastic potential function and the yield function have the same form, is used. Therefore, the rate of plastic strain can be expressed as follows:

$$\dot{\boldsymbol{\epsilon}}^p = \dot{\phi} \frac{\partial f}{\partial \boldsymbol{\sigma}} \quad (20)$$

where the expression for the gradient of the yield function is:

$$\frac{\partial f}{\partial \boldsymbol{\sigma}} = \frac{3}{M^2} \mathbf{s} + \frac{(2p - p_c)}{3} \mathbf{1} \quad (21)$$

3.1.4. Hardening law

The plastic volume strain is used as the hardening parameter in the MCC model to record the loading history of the soil. The following hardening law in rate form can be obtained from the isotropic compression test of soil:

$$\dot{p}_c = c_p p_c \dot{\epsilon}_v^p \quad (22)$$

where $c_p = (1 + e_0) / (\lambda - \kappa)$. λ is the compression index, and

$$\dot{\epsilon}_v^p = \frac{\partial f}{\partial p} \dot{\phi} = (2p - p_c) \dot{\phi} \quad (23)$$

3.2. Integral scheme

In the previous section, the MCC model was presented in rate form. For the stress updating procedure, the rate constitutive equations need to be discretized in time. By utilizing the backward Euler scheme, the integral scheme of stress can be obtained by integrating Eq. (1).

$$\boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}_n + \bar{\mathbf{D}}(\bar{K}, \bar{G}) : (\Delta \boldsymbol{\epsilon}_{n+1} - \Delta \boldsymbol{\epsilon}_{n+1}^p) \quad (24)$$

where the secant bulk modulus $\bar{K}$ and secant shear modulus $\bar{G}$ are used [7]. The expression of $\bar{K}$ is

$$\bar{K} = \frac{p_n}{\Delta \epsilon_{v,n+1}^e} [\exp(c_k \Delta \epsilon_{v,n+1}^e) - 1] \quad (25)$$

where $\Delta \epsilon_{v,n+1}^e = \Delta \epsilon_{v,n+1} - \Delta \epsilon_{v,n+1}^p = \Delta \epsilon_{v,n+1} - \Delta \phi_{n+1} (2p_{n+1} - p_{c,n+1})$. Referring to Borja's practice [7], the integral formula of $\boldsymbol{\sigma}$ can be replaced by those of p and q . This practice can effectively reduce the number of nonlinear equations in the iterative calculation process. The following expressions can be obtained by evaluating the volumetric and deviatoric parts of $\boldsymbol{\sigma}_{n+1}$:

$$p_{n+1} = \frac{1}{3} \boldsymbol{\sigma}_{n+1} : \mathbf{1} = p_n \exp \{c_k [\Delta \epsilon_{v,n+1} - \Delta \phi_{n+1} (2p_{n+1} - p_{c,n+1})]\} \quad (26)$$

$$q_{n+1} = \sqrt{\frac{3}{2}} \|\mathbf{s}_{n+1}\| \quad (27)$$

where the deviatoric stress tensor $\mathbf{s}_{n+1}$ is determined by:

$$\mathbf{s}_{n+1} = \mathbf{s}_n + 2\bar{G}(\Delta\boldsymbol{\gamma}_{n+1} - \Delta\boldsymbol{\gamma}_{n+1}^p) \quad (28)$$

where $\Delta\boldsymbol{\gamma}_{n+1}$ and $\Delta\boldsymbol{\gamma}_{n+1}^p$ are the total and plastic deviatoric strain increment tensors, respectively. $\Delta\boldsymbol{\gamma}_{n+1}^p$ is determined by:

$$\Delta\boldsymbol{\gamma}_{n+1}^p = \frac{\partial f}{\partial \mathbf{s}_{n+1}} \Delta\phi_{n+1} = \frac{3\Delta\phi_{n+1}\mathbf{s}_{n+1}}{M^2} \quad (29)$$

Substituting Eq. (29) into Eq. (28) and reorganizing the new equation, we can obtain:

$$\mathbf{s}_{n+1} = \frac{\mathbf{s}_n + 2\bar{G}\Delta\boldsymbol{\gamma}_{n+1}}{1 + 6\bar{G}\Delta\phi_{n+1}/M^2} \quad (30)$$

Substituting Eq. (30) into Eq. (27), the expression of q_{n+1} can be rewritten as follows:

$$q_{n+1} = \sqrt{\frac{3}{2}} \|\mathbf{s}_{n+1}\| = \sqrt{\frac{3}{2}} \frac{\|\mathbf{s}_n + 2\bar{G}\Delta\boldsymbol{\gamma}_{n+1}\|}{1 + 6\bar{G}\Delta\phi_{n+1}/M^2} \quad (31)$$

Once p_{n+1} and q_{n+1} are determined, $\boldsymbol{\sigma}_{n+1}$ can be obtained as follows:

$$\boldsymbol{\sigma}_{n+1} = p_{n+1}\mathbf{1} + \mathbf{s}_{n+1} \quad (32)$$

The evaluation of $p_{c,n+1}$ can be obtained by integrating Eq. (22):

$$p_{c,n+1} = p_{c,n} \exp[c_p \Delta\phi_{n+1} (2p_{n+1} - p_{c,n+1})] \quad (33)$$

Combining the *FB* smoothing function composed of the yield function f_{n+1} and the plastic multiplier $\Delta\phi_{n+1}$, the closed control equations $\{\mathbf{f}(\mathbf{x})\}_{n+1} = \{f_1 \ f_2 \ f_3 \ f_4\}_{n+1}^T$ can be obtained as follows:

$$\begin{Bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{Bmatrix}_{n+1} = \begin{Bmatrix} c_d \{p_{n+1} - p_n \exp[c_k \Delta\varepsilon_{v,n+1} - c_k \Delta\phi_{n+1} (2p_{n+1} - p_{c,n+1})]\} \\ c_d \left(q_{n+1} - \sqrt{\frac{3}{2}} \frac{\|\mathbf{s}_n + 2\bar{G}\Delta\boldsymbol{\gamma}_{n+1}\|}{1 + 6\bar{G}\Delta\phi_{n+1}/M^2} \right) \\ c_d \{p_{c,n+1} - p_{c,n} \exp[c_p \Delta\phi_{n+1} (2p_{n+1} - p_{c,n+1})]\} \\ \sqrt{(c_d \Delta\phi_{n+1})^2 + f_{n+1}^2} + 2\beta - c_d \Delta\phi_{n+1} + f_{n+1} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \quad (34)$$

where $f_{n+1} = q_{n+1}^2/M^2 + p_{n+1}(p_{n+1} - p_c)$. There are four independent variables $\{\mathbf{x}_{n+1}\} = \{p_{n+1} \ q_{n+1} \ p_{c,n+1} \ \Delta\phi_{n+1}\}^T$ in Eq. (34). It is recommended that $c_d = \|\boldsymbol{\sigma}_n + \bar{\mathbf{D}} : \Delta\boldsymbol{\varepsilon}_{n+1}\|$ out of consideration the magnitude difference of residual errors between $f_{1\sim 3}$ and f_4 . There are also other forms of smooth functions, e.g., *Chen-Mangasarian* and *Chen-Harker-Kanzow-Smale* smooth functions [19]. The former employs an approximation form with exponential function, and the function form of the latter is similar to that of the *FB* function. Both of them can reasonably approximate the *KT* conditions. However, it is easy to suffer from a floating-point overflow problem when the yield function is used as a variable in the exponential function, at least for the *MCC* model. Therefore, the *FB* function is used in this paper. A more detailed discussion on the different smooth functions can be found in the literature [19].

4. Implementation of the MCC model based on the non-monotonic trust region method

The problem with nonlinear equations in Eq. (34) is equivalent to an unconstrained minimization problem, which corresponds to a merit function constructed by Eq. (34) and written as follows:

$$\min \quad \psi(\{\mathbf{x}\}_{n+1}) = \frac{1}{2} \{f(\mathbf{x})\}_{n+1}^T \{f(\mathbf{x})\}_{n+1} \quad (35)$$

where solving the nonlinear equations is equivalent to minimizing the merit function. For convenience, the subscript $n+1$ is omitted in the remainder of this section. The solution of Eq. (35) can be obtained by performing the following

iteration:

$$\{\mathbf{x}\}^{k+1} = \{\mathbf{x}\}^k + \alpha \{\mathbf{d}\} \quad (36)$$

where α and $\{\mathbf{d}\}$ are the step size and variables direction vector, respectively. In the trust region method, the step size is limited first by the trust region radius. Then, the search direction $\{\mathbf{d}\}$ is determined by the descent direction of a model function that is the approximation of the original merit function in a region around the current point. Generally, the trust region method uses a quadratic model to approximate the original merit function as follows:

$$\begin{aligned} \min \quad m^k(\{\mathbf{d}\}) &= \psi^k + \{\mathbf{g}\}_k^T \{\mathbf{d}\} + \frac{1}{2} \{\mathbf{d}\}^T [\mathbf{B}]^k \{\mathbf{d}\} \\ \text{s.t.} \quad \|\{\mathbf{d}\}\| &\leq \Delta^k \end{aligned} \quad (37)$$

where $\{\mathbf{g}\}^k$ is the gradient of ψ^k and $\{\mathbf{g}\}^k = \{\mathbf{f}\}_k^T \{\nabla \mathbf{f}\}^k$. The details of the Jacobian matrix $\{\nabla \mathbf{f}\}$ corresponding to Eq. (34) are presented in Appendix A. $[\mathbf{B}]^k = \{\nabla \mathbf{f}\}_k^T \{\nabla \mathbf{f}\}^k$ is the Hessian matrix. Δ^k is the region radius, and $\Delta^k \leq \Delta_{\max}$. Eq. (37) is called a trust region subproblem, where the variable vector $\{\mathbf{d}\}$ can be determined by minimizing the quadratic function m^k in the trust region. To ensure that the trust region method is still effective in dealing with an optimization problem that has poor scaling, Eq. (37) is usually recast by a scaling matrix as follows:

$$\begin{aligned} \min \quad m^k(\{\tilde{\mathbf{d}}\}) &= \psi^k + \{\tilde{\mathbf{g}}\}_k^T \{\tilde{\mathbf{d}}\} + \frac{1}{2} \{\tilde{\mathbf{d}}\}^T [\tilde{\mathbf{B}}]^k \{\tilde{\mathbf{d}}\} \\ \text{s.t.} \quad \|\{\tilde{\mathbf{d}}\}\| &\leq \Delta^k \end{aligned} \quad (38)$$

where $\{\tilde{\mathbf{g}}\}^k = [\mathbf{D}^s]^{-1} \{\mathbf{g}\}^k$, $\{\tilde{\mathbf{d}}\}^k = [\mathbf{D}^s] \{\mathbf{d}\}^k$, and $[\tilde{\mathbf{B}}]^k = [\mathbf{D}^s]^{-1} [\mathbf{B}]^k [\mathbf{D}^s]^{-1}$. $[\mathbf{D}^s]$ is a diagonal scaling matrix. In the literature [36,37], it is recommended to set the scaling matrix to $[\mathbf{D}_{ii}^s] = [\mathbf{B}_{ii}]^0$. This can be interpreted as follows: if B_{ii}^0 is large, then it is likely that the merit function ψ is more sensitive to changes in certain components x_i . The approximation of m to ψ is questionable, and thus, the search area should be restricted along the i th coordinate. In addition, the advantage of this approach is that there is no need to customize the elements of the diagonal scaling matrix according to specific physical problems, so it is universal than other methods. The poor scaling problems reported in the literature [30,33] can also be avoided when the trust region method is applied to stress updating.

Once the trust region subproblem in Eq. (38) is solved, the quality of the trial step $\{\mathbf{x}^*\}^{k+1} = \{\mathbf{x}\}^k + \{\mathbf{d}\}$ can be evaluated by the following ratio:

$$\rho^k = \frac{\psi(\{\mathbf{x}\}^k) - \psi(\{\mathbf{x}\}^k + \{\mathbf{d}\})}{m_k(\{\mathbf{0}\}) - m_k(\{\mathbf{d}\})} \quad (39)$$

where the numerator denotes the actual reduction of the merit function and the denominator denotes the predicted reduction of the model function. If ρ^k is close to 1, this shows that the model function and the merit function are very good approximations in the trust region. The trial point $\{\mathbf{x}^*\}^{k+1}$ can be accepted as the next iteration point $\{\mathbf{x}\}^{k+1}$, and the trust region radius Δ^k is amplified for the next iterative calculation. If ρ^k is close to 0 or even less than 0, then one should consider rejecting the trial step and reducing the trust region radius to resolve the trust region subproblem in Eq. (38). It is observed from the basic idea of the trust region method that three areas need to be addressed for the successful application of the trust region method: (i) How can the trust region radius be determined? (ii) What are the criteria for accepting the trial point? (iii) How can the trust region subproblem be solved?

4.1. Determination of the trust region radius

The reasonable choice of the trust region radius, including the initial trust region radius Δ^0 and the maximum trust region radius $\Delta_{\max}$, has a great influence on the convergence speed and stability of the trust region method [52]. Utilizing too small a trust region radius reduces the convergence speed. If the trust region radius is too large, it will lose meaning. The reliability of the approximation of the model function to the merit function also decreases. In numerous studies regarding the trust region method that do not involve practical physical problems [29,53,54], there is little evidence to support a particular setting of the trust region radius, or additional calculation efforts are required to provide a reasonable constraint range for the trust region radius. However, for the elastoplastic stress updating problem based on an incremental calculation, the maximum admissible stress increment can be determined

by the difference between the converged stresses of step n and the elastic trial stress of step $n + 1$. The upper limit of the allowable plastic strain increment is the total strain increment. Therefore, the maximum change among the four variables in Eq. (34) can be determined as follows:

$$\{\Delta \mathbf{x}\}_{\max} = \begin{pmatrix} \frac{p_n \exp(c_k \Delta \varepsilon_{v,n+1}) - p_n}{\sqrt{\frac{3}{2}} \|\mathbf{s}_n + 2\bar{G} \Delta \boldsymbol{\gamma}_{n+1}\| - q_n} \\ \frac{p_{c,n} \exp(c_p \Delta \varepsilon_{v,n+1}) - p_{c,n}}{\sqrt{\frac{\Delta \varepsilon_{v,n+1}^2 + 2 \|\Delta \boldsymbol{\gamma}_{n+1}\|^2 / 3}{(2p_n - p_{c,n})^2 + (2q_n/M^2)}}} \end{pmatrix} \quad (40)$$

and considering the influence of the diagonal scaling matrix on the trust region radius, the upper limit of the trust region radius can be determined by:

$$\Delta_{\max} = \|\mathbf{D}^s\| \{\Delta \mathbf{x}\}_{\max} \quad (41)$$

The initial trust region radius Δ^0 can be set to $\Delta_{\max}$ to consider the situation involving single-step iterative convergence.

4.2. Non-monotonic trust region method

In the basic trust region method, the criteria for accepting the trial point is determined by considering only the reduction in the current iteration step. Trial points that cause increases in the merit function are not allowed to be accepted. The basic framework of the trust region method following this monotonic search strategy is detailed below:

Step 0. Initialization. The initial iteration point $\{\mathbf{x}\}^0$, the trust region radius Δ^0 , and the small positive tolerance $FTOL$ are given. $\Delta_{\max}$ is chosen as the upper limit of the trust region radius. Set $k = 0$.

Step 1. Convergence judgment. Compute the norm of the residual vector $\|\mathbf{f}(\{\mathbf{x}\}^k)\|$. If $\|\mathbf{f}(\{\mathbf{x}\}^k)\| \leq FTOL$, the iterative process is terminated.

Step 2. Step calculation. Solve the trust region subproblem in Eq. (38) and obtain the trial point $\{\mathbf{x}^*\}^{k+1}$.

Step 3. Estimation of the trial point. Compute the ratio ρ^k by using Eq. (39).

If $\rho_k > 0.1$, accepted the trial point and set $\{\mathbf{x}\}^{k+1} = \{\mathbf{x}^*\}^{k+1}$; otherwise, set $\{\mathbf{x}\}^{k+1} = \{\mathbf{x}\}^k$.

Step 4. Adjustment of the region radius. Adjust the trust region radius in the next iteration step as follows:

$$\Delta^{k+1} = \begin{cases} 0.5\Delta^k & \rho^k < 0.1 \\ \Delta^k & 0.1 \leq \rho^k \leq 0.75 \\ \min(2\Delta^k, \Delta_{\max}) & \rho^k > 0.75 \end{cases} \quad (42)$$

Set $k = k + 1$ and go to *Step 1*.

Based on the ratio ρ_k of predicted reduction and actual reduction, the quality of the trial step is divided into three kinds: 'bad', 'good', and 'very good'. If the trial step is 'bad', discard it and reduce the trust region radius. If the trial step is 'good' or 'very good', accept it. The difference is that the trust region radius will remain constant for the 'good' case and be enlarged for the 'very good' case. As seen from *Step 3*, however, in the monotonic trust region (MTR) method, the merit function is required to decrease continuously in each successful iteration because the trial point can be accepted only when ρ_k is greater than 0. However, this condition is too harsh for practical physical problems, and it may lead the search to fall into a local optimum and reduce the possibility of finding the optimal solution. Therefore, some researchers try to introduce a non-monotonic strategy in the acceptance criteria for trial points [38,55,56]. For the non-monotonic strategy, the merit function in the search process is allowed a limited increase, but the overall trend needs to remain downward. This treatment can effectively enhance the robustness of the trust region method with very little extra calculation cost. In this paper, the non-monotonic strategy proposed by Toint [38] is used to improve the monotonic trust region method above, where only *Step 3* needs to be adjusted.

Step 3 of the non-monotonic trust region method. Estimation of the trial point. Compute the largest index such that

$$\psi^{M(k)} = \max_{i=\max[l-M_{\max}, 0], \dots, l} \psi^i \quad (43)$$

where $\psi^i = \psi(\{\mathbf{x}\}^i)$ is the sequence of values of the merit function at the ends of successful iterations. $M_{\max}$ is a positive integer indicating the maximum number of previously successful iterations that are considered when evaluating the trial point. It is recommended to set $M_{\max}$ to 10 [38]. l denotes the number of successful iterations and is initialized to 0 in Step 0.

Then, compute $\psi(\{\mathbf{x}\}^k + \{\mathbf{d}\})$ and the following ratio:

$$\rho_1^k = \begin{cases} \frac{\psi^{M(k)} - \psi(\{\mathbf{x}\}^k + \{\mathbf{d}\})}{\sum_{i=M(k)}^l d^i} & \text{if } l > 0, \\ 0 & \text{otherwise} \end{cases} \quad (44)$$

Compute the ratio of the actual reduction to the predicted reduction at the current step:

$$\rho_2^k = \frac{\psi(\{\mathbf{x}\}^k) - \psi(\{\mathbf{x}\}^k + \{\mathbf{d}\})}{m_k(\{\mathbf{0}\}) - m_k(\{\mathbf{d}\})} \quad (45)$$

Finally, the ratio used to evaluate the trial point is determined by:

$$\rho^k = \max(\rho_1^k, \rho_2^k) \quad (46)$$

If $\rho^k < 0.1$, then set $\{\mathbf{x}\}^{k+1} = \{\mathbf{x}\}^k$. Otherwise, set $\{\mathbf{x}\}^{k+1} = \{\mathbf{x}^*\}^{k+1}$, increment l by one and set

$$\psi^l = \psi(\{\mathbf{x}\}^k + \{\mathbf{d}\}) \text{ and } d^l = m_k(\{\mathbf{0}\}) - m_k(\{\mathbf{d}\}) \quad (47)$$

4.3. Double dogleg method for solving the trust region subproblem

Methods for solving trust region subproblems have been widely studied [57], such as the singular dogleg method [58], double dogleg method [59], two-dimensional subspace minimization method [60], and conjugate gradient method [61]. Herein, the double dogleg method is used to approximate the solution of the trust region subproblem. The double dogleg method is an improved version of the single dogleg method. In both methods, the dogleg step is determined by a combination of the Cauchy step and Newton step. According to the relative positions of the Cauchy point and the Gauss–Newton point to the trust region boundary, three cases for the determination of the dogleg step are schematically represented in Fig. 5.

First, the Cauchy point $\{\mathbf{x}_C\}^{k+1} = \{\mathbf{x}\}^k + \{\mathbf{d}_C\}$ is determined in the steepest descent direction, where $\{\mathbf{d}_C\} = -\{\tilde{\mathbf{g}}\}^k$. Then, the position of the Cauchy point relative to the trust region boundary needs to be estimated. If $\|\{\mathbf{d}_C\}\| \geq \Delta^k$, the intersection point between the Cauchy step and trust region boundary is used as the minimum point of the quadratic model, as shown in Fig. 5(a). If the Cauchy point is interior, we continue to calculate the Newton point $\{\mathbf{x}_{GN}\}^{k+1} = \{\mathbf{x}\}^k + \{\mathbf{d}_{GN}\}$, where $\{\mathbf{d}_{GN}\} = -[\tilde{\mathbf{B}}]_k^{-1} \{\tilde{\mathbf{g}}\}^k$. If the Newton point is interior, as shown in Fig. 5(b), it is used as the minimum point of the quadratic model. If $\|\{\mathbf{d}_{GN}\}\| > \Delta^k$, the dogleg step is calculated, as shown in Fig. 5(c). In the single dogleg method, the minimum point of the quadratic model is determined by the intersection point of between the trust region boundary and the dogleg segment, which is a segment connecting the Cauchy point and Newton point. In the double dogleg method, the position of the point connecting the Cauchy point to form the dogleg segment is moved towards the trust region boundary, as shown in Fig. 5(c). The search direction generated by the double dogleg method is more inclined to the newton direction than that produced by the single dogleg step. Numerical experimental results have shown that the performance of the algorithm can be improved to a certain extent by this treatment [59].

The double dogleg method is detailed below to complete *Step 2* in the non-monotonic trust region method:

Step 2.0. Calculate the gradient and Hessian of the merit function at the current point $\{\mathbf{x}\}^k$ and scale them diagonally as follows:

$$\{\tilde{\mathbf{g}}\}^k = [\mathbf{D}^s]^{-1} \{\mathbf{g}\}^k = [\mathbf{D}^s]^{-1} \{\mathbf{f}\}_k^T [\nabla \mathbf{f}]^k \quad (48)$$

$$[\tilde{\mathbf{B}}]^k = [\mathbf{D}^s]^{-1} [\mathbf{B}]^k [\mathbf{D}^s]^{-1} = [\mathbf{D}^s]^{-1} [\nabla \mathbf{f}]_k^T [\nabla \mathbf{f}]^k [\mathbf{D}^s]^{-1} \quad (49)$$

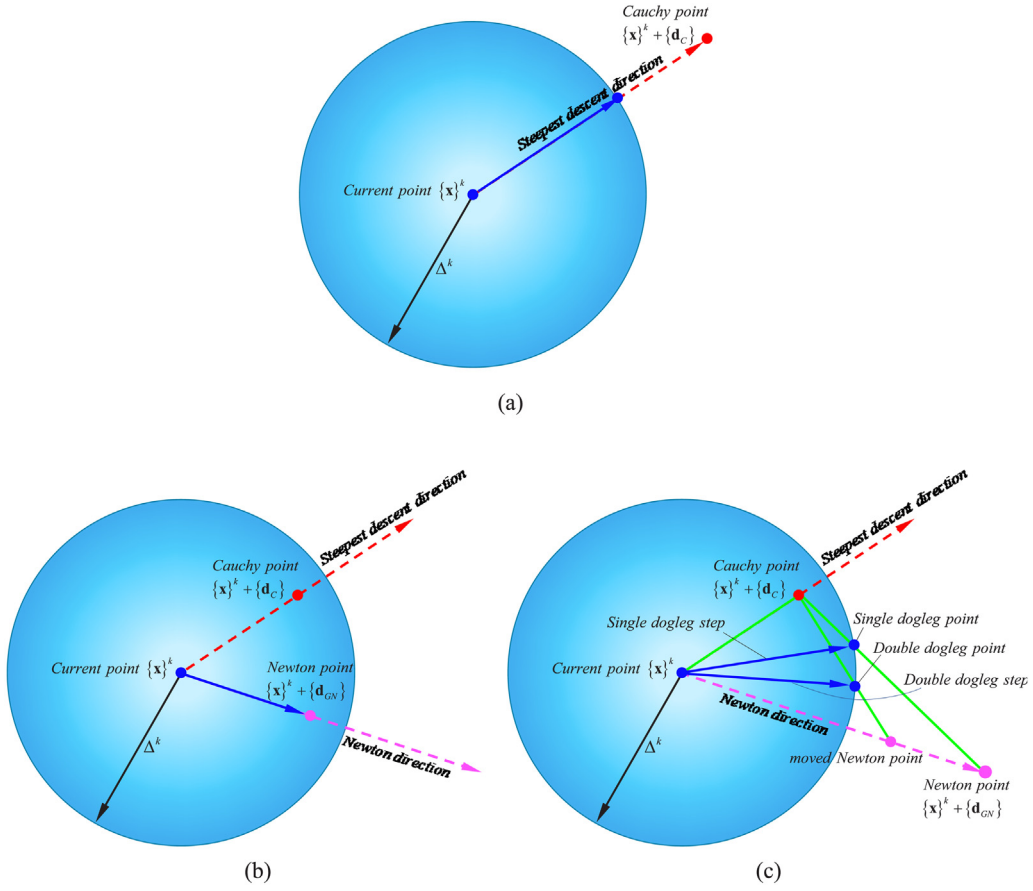


Fig. 5. Three cases of the dogleg method: (a) Cauchy step; (b) Newton step; (c) Single/double dogleg step.

Step 2.1. Calculate the Cauchy step:

$$\{\mathbf{d}_C\} = -\tau_1^k \Delta^k \frac{\{\tilde{\mathbf{g}}\}^k}{\|\{\tilde{\mathbf{g}}\}^k\|} \quad (50)$$

where the coefficient τ_1^k guarantees that the Cauchy step does not exceed the trust region boundary and is expressed as follows:

$$\tau_1^k = \min \left[1, \frac{\|\{\tilde{\mathbf{g}}\}^k\|^3}{\Delta^k \{\tilde{\mathbf{g}}\}_k^T [\tilde{\mathbf{B}}]^k \{\tilde{\mathbf{g}}\}^k} \right] \quad (51)$$

If $\|\{\mathbf{d}_C\}\| = \Delta^k$, set $\{\mathbf{d}\} = [\mathbf{D}^s]^{-1} \{\mathbf{d}_C\}$ and go to Step 2.4.

Step 2.2. If $\|\{\mathbf{d}_C\}^k\| < \Delta^k$, then calculate the Newton step:

$$\{\mathbf{d}_{GN}\} = -[\tilde{\mathbf{B}}]_k^{-1} \{\tilde{\mathbf{g}}\}^k \quad (52)$$

If $\|\{\mathbf{d}_{GN}\}\| \leq \Delta^k$, set $\{\mathbf{d}\} = [\mathbf{D}^s]^{-1} \{\mathbf{d}_{GN}\}$ and go to Step 2.4.

Step 2.3. If $\|\{\mathbf{d}_{GN}\}^k\| > \Delta^k$, then calculate the double dogleg step:

$$\{\mathbf{d}_D\} = \{\mathbf{d}_C\} + \tau_2^k (\tau_3^k \{\mathbf{d}_{GN}\} - \{\mathbf{d}_C\}) \quad (53)$$

where $\tau_3^k \in (\tau_4^k, 1)$ and τ_4^k is obtained as follows:

$$\tau_4^k = \frac{\|\{\tilde{\mathbf{g}}\}^k\|^4}{\left(\{\tilde{\mathbf{g}}\}_k^T [\tilde{\mathbf{B}}]^k \{\tilde{\mathbf{g}}\}^k\right) \left(\{\tilde{\mathbf{g}}\}_k^T [\tilde{\mathbf{B}}]_k^{-1} \{\tilde{\mathbf{g}}\}^k\right)} \quad (54)$$

where τ_3^k is usually set to $0.8\tau_4^k + 0.2$. The double dogleg method degenerates into the single dogleg method under the condition where $\tau_3^k = 1$. τ_2^k can be obtained by solving the quadratic equation $\|\{\mathbf{d}_C\} + \tau_2^k (\tau_3^k \{\mathbf{d}_{GN}\} - \{\mathbf{d}_C\})\| = \Delta_k$ as follows:

$$\tau_2^k = \frac{-\{\mathbf{v}\}_k^T \{\mathbf{d}_C\} \pm \sqrt{(\{\mathbf{v}\}_k^T \{\mathbf{d}_C\})^2 - \|\{\mathbf{v}\}^k\|^2 (\|\{\mathbf{d}_C\}\|^2 - \Delta_k^2)}}{\|\{\mathbf{v}\}^k\|^2} \quad (55)$$

where $\{\mathbf{v}\}^k = \tau_3^k \{\mathbf{d}_{GN}\} - \{\mathbf{d}_C\}$, and the positive root of Eq. (55) is always chosen due to $0 \leq \tau_2^k \leq 1$. Finally, set $\{\mathbf{d}\} = [\mathbf{D}^*]^{-1} \{\mathbf{d}_D\}$.

Step 2.4. Exit with $\{\mathbf{x}^*\}^{k+1} = \{\mathbf{x}\}^k + \{\mathbf{d}\}$.

The implementation of the MCC model based on the proposed algorithm is briefly summarized as follows: First, the rate constitutive equations of the MCC model constrained by the loading/unloading equality are discretized into a set of nonlinear algebraic equations by the backward Euler scheme. Then, the nonlinear equations constrained by the loading/unloading equality are transformed into a smooth form by using the *FB* smoothing function instead of the non-smooth *KT* conditions. Finally, the smooth nonlinear constitutive equations are considered as an equivalent unconstrained minimization problem, which is solved further by the double dogleg trust region method (improved by the non-monotonic strategy). The computation procedures of the proposed stress updating algorithm for the MCC model are presented in Fig. 6. It should be emphasized that the proposed algorithm can work for all elastoplastic models. For the numerical implementation of soil models based on the critical state concept, there are typically nonlinear and non-smooth characteristics, which can be addressed better by the proposed algorithm. In addition, the smooth function and trust region method can work for the finite strain plasticity [62,63] because they are essentially used to address common computational difficulties in the numerical implementation of the plastic model, i.e., the nonsmoothness and nonlinearity. In this paper, the MCC model with the small formulation is used as an example.

5. Consistent and continuum tangent operators

Under the framework of solid mechanics, a nonlinear boundary value problem can be solved by the finite element method. The external load acting on the given structure is applied incrementally. The change in the internal force comes from the updates of the state variables at the material point. The unbalanced forces of the element nodes are minimized iteratively by the newton method so that the structure transitions from the previous equilibrium state n to the next equilibrium state $n + 1$. When the internal force and external force reach equilibrium on each element node, the following equation holds:

$$\{\mathbf{F}(\boldsymbol{\sigma})\}_{n+1} = \{\mathbf{F}^{\text{ext}}\}_{n+1} - \{\mathbf{F}^{\text{int}}\}_{n+1} = \{\mathbf{0}\} \quad (56)$$

where $\{\mathbf{F}\}_{n+1}$, $\{\mathbf{F}^{\text{int}}\}_{n+1}$, and $\{\mathbf{F}^{\text{ext}}\}_{n+1}$ denote the global residual force vector, the global internal force vector, and the global external force vector, respectively. By utilizing the displacement-based finite method with $\boldsymbol{\sigma}_{n+1}^i = \boldsymbol{\sigma}(\boldsymbol{\epsilon}_{n+1}^i) = \boldsymbol{\sigma}(\boldsymbol{\epsilon}(\mathbf{U}_{n+1}^i))$, Eq. (56) is solved by the following linearized form:

$$[\mathbf{K}]_{n+1}^i \{\delta \mathbf{U}\}_{n+1}^{i+1} = - \left(\{\mathbf{F}^{\text{ext}}\}_{n+1} - \{\mathbf{F}^{\text{int}}\}_{n+1}^i \right) \quad (57)$$

where $\{\delta \mathbf{U}\}_{n+1}^{i+1}$ is a displacement correction. The global stiffness matrix $[\mathbf{K}]_{n+1}^i$ can be determined by:

$$[\mathbf{K}]_{n+1}^i = \begin{bmatrix} \frac{\partial \mathbf{F}}{\partial \boldsymbol{\sigma}} \end{bmatrix}_{n+1}^i \begin{bmatrix} \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\epsilon}} \end{bmatrix}_{n+1}^i \begin{bmatrix} \frac{\partial \boldsymbol{\epsilon}}{\partial \mathbf{U}} \end{bmatrix}_{n+1}^i \quad (58)$$

where $[\partial \boldsymbol{\sigma} / \partial \boldsymbol{\epsilon}]_{n+1}^i$ is determined by the constitutive equations at the material point scale. $[\partial \boldsymbol{\sigma} / \partial \boldsymbol{\epsilon}]_{n+1}^i$ is called the CTO [41]. As a spare option, the CON is sometimes used for some highly complex constitutive models when the determination of the second derivatives of the yield function and plastic potential function is exceedingly laborious.

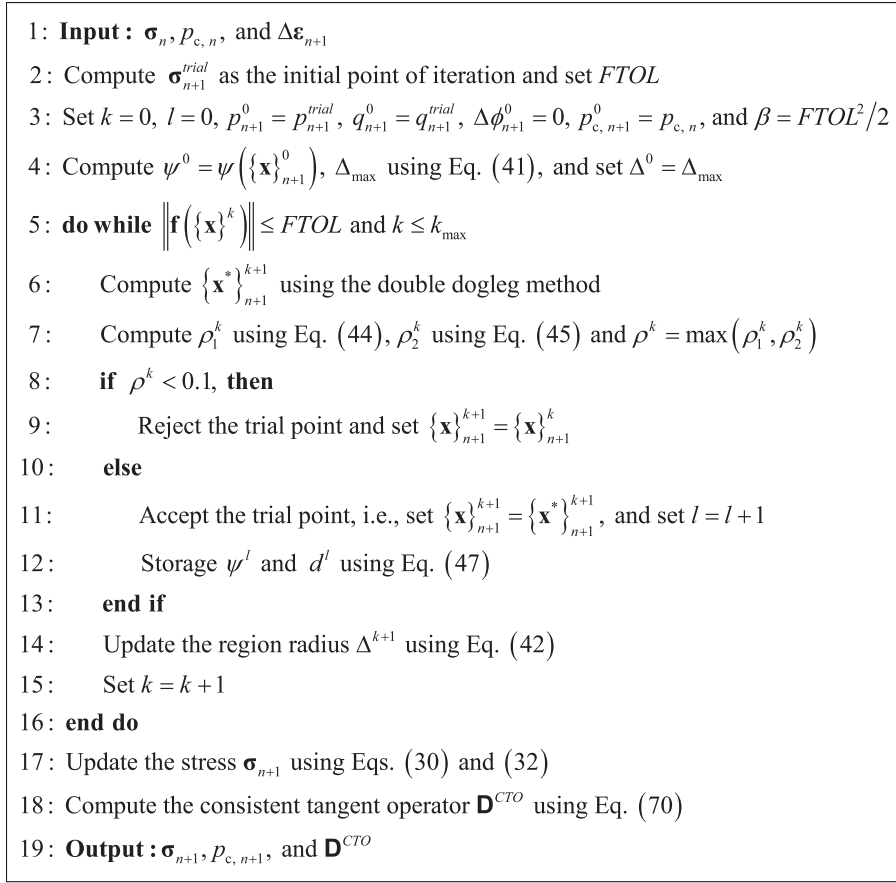


Fig. 6. Flow chart of the stress updating algorithm.

However, it has been reported that the use of the CON affects the quadratic convergence speed of the global solution and requires a much higher calculation cost. In what follows, the algebraic and geometric interpretations of the CTO and CON are discussed first by a simple descriptive example that is analogous to Eq. (56). Then, the CTO and CON of the MCC model, which is consistent with the unconstrained stress updating scheme, are derived.

5.1. Algebraic and geometric interpretations

We specify a simple vector function $\{\mathbf{F}\} = \{\mathbf{F}^{ext}\} - \{g_1^{0.5} \ g_2^{0.8}\}^T = \{\mathbf{0}\}$ as an analog of the residual force equation, in which $\{g_1^{0.5} \ g_2^{0.8}\}^T$ and $\{\mathbf{g}\}$ are used as the analog of $\{\mathbf{F}^{int}\}$ and $\{\sigma\}$ in Eq. (56), respectively. The constitutive equation used to evaluate $\{\mathbf{g}\}$ is defined by the following differential form:

$$\{d\mathbf{g}\} = [\mathbf{D}(\mathbf{x})] \{d\mathbf{x}\} \quad (59)$$

where $\{\mathbf{x}\}$ is used as an analog of the strain $\{\epsilon\}$ (or displacement $\{\mathbf{U}\}$). $[\mathbf{D}(\mathbf{x})]$ can be called the CON and is defined in this descriptive example as follows:

$$[\mathbf{D}(\mathbf{x})] = \begin{bmatrix} 1 + x_1 + x_2 & 0.2 + x_1 + x_2 \\ 0.2 + x_1 + x_2 & 1 + x_1 + x_2 \end{bmatrix} \quad (60)$$

If the state variables in the previous step are known, for the given $\{\Delta \mathbf{x}\}_{n+1}^i$, $\{\mathbf{g}\}_{n+1}^i$ can be determined based on the backward Euler scheme as follows:

$$\{\mathbf{g}\}_{n+1}^i = [\mathbf{D}(\mathbf{x})]_{n+1}^i (\{\mathbf{x}\}_{n+1}^i - \{\mathbf{x}\}_n) + \{\mathbf{g}\}_n \quad (61)$$

To facilitate the description of the differences and relations between the two tangent operators, the theoretical solution of Eq. (59) (usually unknown) can be given as follows due to its simplicity:

$$\{\mathbf{g}\} = \begin{Bmatrix} x_1 + 0.2x_2 + 0.5(x_1 + x_2)^2 \\ 0.2x_1 + x_2 + 0.5(x_1 + x_2)^2 \end{Bmatrix} \quad (62)$$

Then, the strain (displacement) correction can be obtained by:

$$\{\delta \mathbf{x}\}_{n+1}^i = -([\mathbf{K}]_{n+1}^i)^{-1} \{\mathbf{F}\}_{n+1}^i \quad (63)$$

where $[\mathbf{K}]_{n+1}^i = [\partial \mathbf{F} / \partial \mathbf{g}]_{n+1}^i [\partial \mathbf{g} / \partial \mathbf{x}]_{n+1}^i$. The expression of $[\partial \mathbf{F} / \partial \mathbf{g}]_{n+1}^i$ is:

$$\left[\frac{\partial \mathbf{F}}{\partial \mathbf{g}} \right]_{n+1}^i = - \begin{bmatrix} 0.5g_1^{-0.5} & 0 \\ 0 & 0.8g_1^{-0.2} \end{bmatrix}_{1,n+1}^i \quad (64)$$

and the CTO of this example is:

$$\begin{aligned} \left[\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right]_{n+1}^i &= [\mathbf{D}(\mathbf{x})]_{n+1}^i + \left[\frac{\partial \mathbf{D}(\mathbf{x})}{\partial \mathbf{x}} \Delta \mathbf{x} \right]_{n+1}^i = \begin{bmatrix} 1 + x_1 + x_2 & 0.2 + x_1 + x_2 \\ 0.2 + x_1 + x_2 & 1 + x_1 + x_2 \end{bmatrix}_{n+1}^i \\ &\quad + (x_{1,n+1}^i + x_{2,n+1}^i - x_{1,n+1} - x_{2,n+1}) \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \end{aligned} \quad (65)$$

It is easy to see the differences and relations between the CTO ($[\partial \mathbf{g} / \partial \mathbf{x}]_{n+1}^i$) and CON ($[\mathbf{D}(\mathbf{x})]_{n+1}^i$) from Eq. (65). From the perspective of algebra, the CTO is the partial derivative of the numerical expression of stress (i.e., Eq. (61)) obtained by the backward Euler scheme to the strain, whereas the CON is the partial derivative of the theoretical expression of stress (i.e., Eq. (62)) to the strain. The difference between the two tangent operators comes from the approximation of the integral scheme to the constitutive equation in differential (or rate) form, and this difference increases with increasing step size. In turn, the CTO degenerates to the CON when the step size tends to zero.

In what follows, the influence of the CTO and CON on the convergence speed and stability of the global solution is explored with different step sizes. The state variables of the previous equilibrium state are set to $\{\mathbf{x}\}_n = \{0.1, 0.2\}^T$, $\{\mathbf{g}\}_n = \{0.23, 0.31\}^T$, $\{\mathbf{F}^{\text{int}}\}_n = \{\mathbf{F}^{\text{ext}}\}_n = \{0.23^{0.5}, 0.31^{0.8}\}^T$ and $\{\mathbf{F}\}_n = \{0, 0\}^T$. The theoretical solution regarding the global external force $\{\mathbf{F}^{\text{ext}}\}_{n+1}$ in the next equilibrium state can be obtained by substituting $\{\Delta \mathbf{x}\}$ into Eq. (61) and solving the force residual equation $\{\mathbf{F}\}_{n+1} = \{\mathbf{0}\}$. Then, the obtained theoretical solution of $\{\mathbf{F}^{\text{ext}}\}_{n+1}$ is used again as the applied external load. Fig. 7 shows the search paths with the two tangent operators and the corresponding convergence maps, where the search behaviour is terminated once $\|\{\mathbf{F}\}_{n+1}\| \leq 10^{-3}$.

The results of the figures demonstrate that the term $[(\partial \mathbf{D}(\mathbf{x}) / \partial \mathbf{x}) \Delta \mathbf{x}]_{n+1}^i$ that causes differences between the CTO and CON has little influence on the search direction when the step size is small. The search paths and iterations based on the CON are similar to those based on the CTO, as shown in Fig. 7(a). With increasing step size, the search direction based on the CON gradually deviates from the Newton direction. More iterations are required compared to the same process with the CTO due to the lost quadratic convergence speed of the newton method, as shown in Fig. 7(b) and (c). When $\{\Delta \mathbf{x}\} = \{1.6, 1.6\}^T$, the iterative search process with the CTO still converges to a pre-defined value, but the iterative search with the CON diverges, as shown in Fig. 7(d). The reason for this is that the initial point of the iteration is outside the convergence radius of the “newton method” perturbed by the CON. Therefore, in applications, the use of the CON may necessitate rather small time steps to achieve the convergence of the global equilibrium iteration.

5.2. Smoothing consistent and continuum tangent operators

In the computational framework of the operator splitting technique, the expressions of the CTO and CON needed to be determined by the different constitutive equations (the elastic equation or plastic equation) based on the

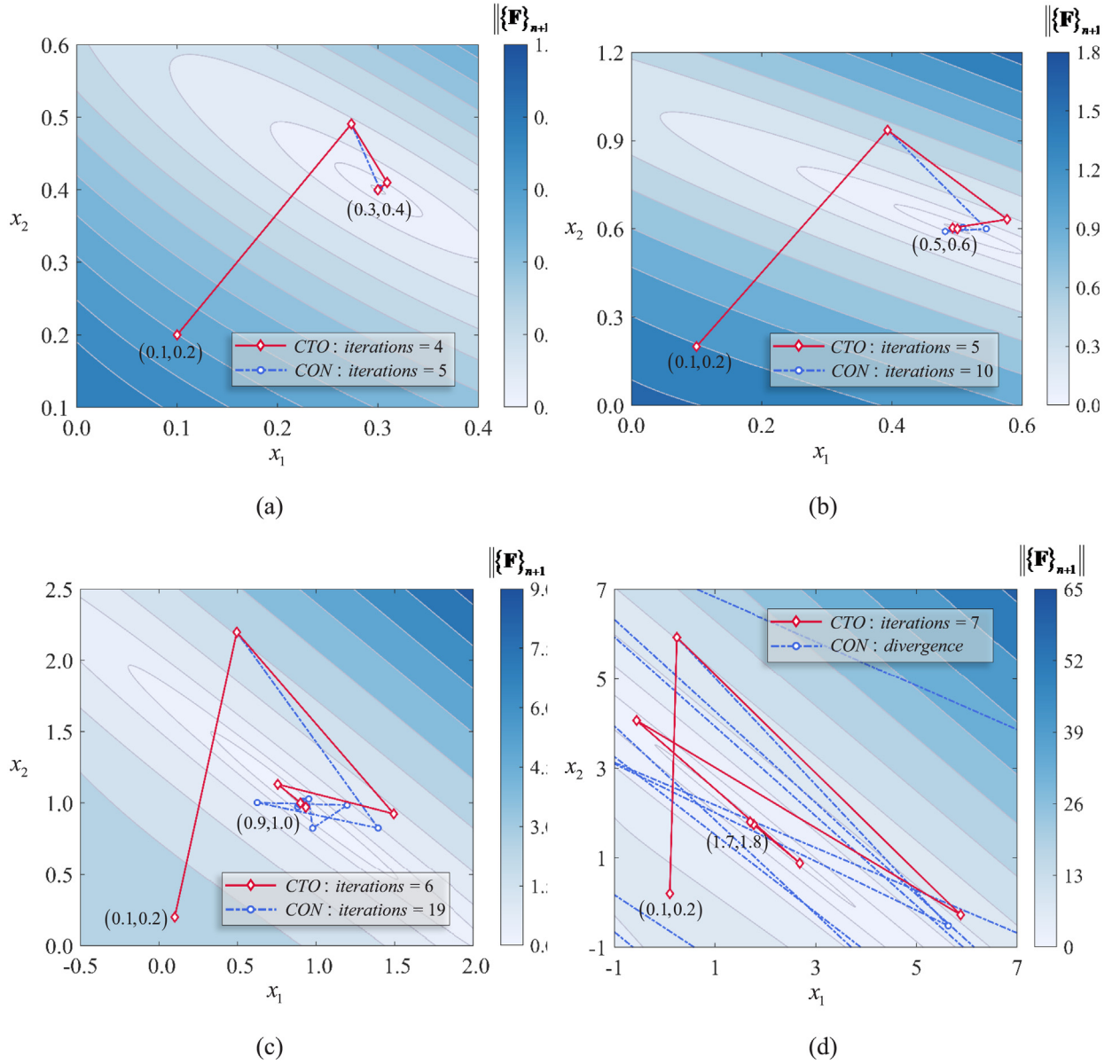


Fig. 7. Search paths for global iteration: (a) $\{\Delta \mathbf{x}\} = \{0.2, 0.2\}^T$; (b) $\{\Delta \mathbf{x}\} = \{0.4, 0.4\}^T$; (c) $\{\Delta \mathbf{x}\} = \{0.8, 0.8\}^T$; (d) $\{\Delta \mathbf{x}\} = \{1.6, 1.6\}^T$.

loading/unloading states of the material at the current incremental step. For the stress updating algorithm presented in this paper, the two tangent operators can be derived in a unified framework due to the use of a smoothing function. Taking the total differential of Eq. (34)₄, we can obtain:

$$\chi_0 \left(\frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} : d\boldsymbol{\sigma}_{n+1} + \frac{\partial f_{n+1}}{\partial p_{c,n+1}} \frac{\partial p_{c,n+1}}{\partial \Delta \phi_{n+1}} d\Delta \phi_{n+1} \right) + \chi_1 d\Delta \phi_{n+1} = 0 \quad (66)$$

where $\chi_0 = f_{n+1} / \sqrt{(c_d \Delta \phi_{n+1})^2 + f_{n+1}^2 + 2\beta + 1}$ and $\chi_1 = c_d^2 \Delta \phi_{n+1} / \sqrt{(c_d \Delta \phi_{n+1})^2 + f_{n+1}^2 + 2\beta} - c_d$ for convenience when writing. $d\boldsymbol{\sigma}_{n+1}$ can be expressed by:

$$d\boldsymbol{\sigma}_{n+1} = \bar{\mathbf{D}} : \left(d\boldsymbol{\varepsilon}_{n+1} - d\Delta \phi_{n+1} \frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} \right) \quad (67)$$

where the expression for $d\Delta\phi_{n+1}$ can be obtained by putting Eq. (67) into Eq. (66) as follows:

$$d\Delta\phi_{n+1} = \frac{\chi_0 \frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} : \bar{\mathbf{D}} : d\boldsymbol{\epsilon}_{n+1}}{\chi_0 \frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} : \bar{\mathbf{D}} : \frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} - \chi_0 \frac{\partial f_{n+1}}{\partial p_{c,n+1}} \frac{\partial p_{c,n+1}}{\partial \Delta\phi_{n+1}} - \chi_1} \quad (68)$$

Substituting Eq. (66) back into Eq. (67) yields the expression of the CON:

$$\mathbf{D}^{CON} = \bar{\mathbf{D}} - \frac{\chi_0 \frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} : \bar{\mathbf{D}} : \frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}}}{\chi_0 \frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} : \bar{\mathbf{D}} : \frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} - \chi_0 \frac{\partial f_{n+1}}{\partial p_{c,n+1}} \frac{\partial p_{c,n+1}}{\partial \Delta\phi_{n+1}} - \chi_1} \quad (69)$$

As discussed in Section 4.1, the CTO of the MCC model based on the unconstrained integral scheme can be obtained by computing the partial derivative of Eq. (24) to the strain $\boldsymbol{\epsilon}_{n+1}$ as follows:

$$\mathbf{D}^{CTO} = \frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = c_0 \mathbf{P} + c_1 \mathbf{1} \otimes \mathbf{1} + c_2 \mathbf{1} \otimes \hat{\mathbf{n}} + c_3 \Delta \boldsymbol{\gamma}_{n+1} \otimes \mathbf{1} + c_4 \Delta \boldsymbol{\gamma}_{n+1} \otimes \hat{\mathbf{n}} + c_5 \hat{\mathbf{n}} \otimes \mathbf{1} + c_6 \hat{\mathbf{n}} \otimes \hat{\mathbf{n}} \quad (70)$$

where the derivation details of Eq. (70) are presented in Appendix B for the readability of the paper. The matrix representation for the fourth-order tensor can be found in Appendix C.

It is noteworthy that there is an interesting property regarding the CTO and CON based on the unconstrained stress updating strategy. Taking the CON as an example, if the current step is elastoplastic loading, then $\Delta\phi_{n+1} > 0$ and $f_{n+1} = 0$. The result by which $\chi_0 = 0 + 1 = 1$ and $\chi_1 = c_d - c_d = 0$ is obtained when β is close to 0. Substituting the conditions $\chi_0 = 1$ and $\chi_1 = 0$ into Eq. (69), Eq. (69) will degenerate into the plastic form of the CON within the computational framework of operator splitting technique as follows:

$$\mathbf{D}^{CON} = \bar{\mathbf{D}} - \frac{\frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} : \bar{\mathbf{D}} : \frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}}}{\frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} : \bar{\mathbf{D}} : \frac{\partial f_{n+1}}{\partial \boldsymbol{\sigma}_{n+1}} - \frac{\partial f_{n+1}}{\partial p_{c,n+1}} \frac{\partial p_{c,n+1}}{\partial \Delta\phi_{n+1}}} \quad (71)$$

Similarly, $\Delta\phi_{n+1} = 0$ and $f_{n+1} < 0$ when the current step is elastic loading. Substituting the corresponding results $\chi_0 = 0$ and $\chi_1 = -c_d$ into Eqs. (69). Eq. (69) can also degenerate into the elastic form of the CON based on the operator splitting technique as follows:

$$\mathbf{D}^{CON} = \bar{\mathbf{D}} \quad (72)$$

This property is also true for the smoothing CTO, which also means that the smoothing CTO is capable of providing a smoothing transition for the description of mechanical behaviour of elastoplastic material even under repeated loading and unloading conditions.

6. Numerical validation

As mature finite-element simulation software, ABAQUS is an excellent analysis platform for assessing algorithm performance. In this section, the MCC model is implemented in ABAQUS based on the proposed unconstrained stress updating algorithm. Then, the performance of the proposed algorithm is analysed and discussed in depth with a series of drained and undrained boundary value problems and is compared with the ABAQUS/Standard default algorithm. As a necessary statement, the following numerical examples are run on a single processor of the same computer. The computer configuration includes an Intel Core i7-9750 processor @ 2.60 GHz, 16 GB of RAM, and the Windows 10 Ultimate 64-bit operating system. The tolerance of the proposed algorithm is set to $FTOL = 10^{-6}$.

6.1. Element test

First, to evaluate the convergence performance of the proposed algorithm in the local calculation, a simple numerical example with only a single element is performed, where the possibility of calculation interruption caused by newton iteration failure at the global scale is minimized. The parameters of the MCC model used for the numerical example are given as follows: $M = 1$, $\lambda = 0.15$, $\kappa = 0.03$, $\nu = 0.3$, and $e_0 = 0.5$. The initial stress state and the pre-consolidation pressure are set to $\sigma_1 = \sigma_2 = \sigma_3 = 200$ kPa and $p_c = 200$ kPa. Note that the $\boldsymbol{\sigma}_n$ is used as the initial point for iteration to create a more demanding iteration condition in this subsection. The loading paths of the element are shown in Fig. 8(a). During the loading process, a condition requiring equal

time step sizes is imposed. Following the unconstrained stress updating strategy presented in Section 2.2, the stress and strain results for the Newton method and the non-monotonic trust region method are presented in Fig. 8(b) to (f). The minimum number of load steps needed to achieve convergence for the two methods are first explored, and then, for the NMTR method, the prediction results under the same steps as that used by the Newton method are also presented. The N denotes the number of the load increment. Smaller N means greater load increment and higher convergence challenges. As expected, the two different implementations give rise to the same results under the same number of load steps, whereas a larger loading increment size is allowed for the NMTR method on the premise of calculation convergence.

Then, the effect of the non-monotonic technique on the trust region method in terms of improvement is presented by two loading and unloading examples. The initial stress state of the numerical examples is represented by the point A in Fig. 9(a). The parameters of the MCC model in the last example are also used here, but the pre-consolidation pressure is set to $p_c = 218$ kPa to ensure that the current yield curve passes through the initial stress point. The strain increments are set to $\Delta\varepsilon_1 = \Delta\varepsilon_2 = -0.015$ and $\Delta\varepsilon_3 = 0$ for the unloading case and to $\Delta\varepsilon_1 = \Delta\varepsilon_2 = 0.03$ and $\Delta\varepsilon_3 = 0$ for the loading case. Only a single load step is considered. Fig. 9(b) and (c) show the changes in $\|\{\mathbf{f}(\mathbf{x})\}_{n+1}\|$ as the number of iterations increases for the MTR method and the NMTR method under loading and unloading cases, respectively. The two methods converge to results satisfying the tolerance requirement under loading conditions, but a distinct computational challenge arises in the unloading case for the MTR method. The reason for this result has been mentioned at the end of Section 2. Namely, for the unloading situation where the initial stress point is on the yield surface, the smoothing function has a very large curvature due to the fact that $f = 0$ and $\Delta\phi = 0$, which easily causes the search behaviour based on the monotonic strategy to fall into a local optimum. A similar phenomenon in which the MTR method may suffer from computational difficulty near areas of high curvature has also been reported in the Ref. [30]. The search ability of the NMTR method is enhanced because the merit function is allowed to increase limitedly at the local scope to achieve a better decline at the global scope. Therefore, the convergence of the algorithm can be guaranteed even under the condition of high curvature. Notably, the computational difficulty of the smoothing function near areas of high curvature can also be solved by choosing an appropriate initial iteration point. For instance, the stress update in the unloading cases can be realized directly without the need for second iterations when the elastic trial point is set to the initial iteration point.

6.2. Collapse of a rigid strip footing

In this subsection, the effectiveness of the smoothing CTO and CON derived by the proposed algorithm is investigated by simulating the collapse of a rigid strip footing, which is a well-known boundary value problem for evaluating the stress updating algorithm used in the constitutive models of soils [8,12,64]. The ABAQUS 6.14/Standard default stress updating algorithm can be used as an objective standard for the validation of the effectiveness of the smoothing CTO since the MCC model has been added to the ABAQUS/Standard Material Library.

The geometry information and element mesh of the footing are shown in Fig. 10(a) and (b), respectively. The element 4 will be used for the following stress path analysis. The material parameters used for the MCC model are obtained from the literature [64] as follows: $M = 0.898$, $\lambda = 0.25$, $\kappa = 0.05$, $\nu = 0.3$, and $e_1 = 1.6$, where e_1 is the critical state void ratio at $p = 1$ kPa. The submerged weight of the soil is set to 6 kN/m^3 . In the numerical simulation, three analysis steps are used, including the geostatic stress analysis step, the unloading analysis step, and the loading analysis step. In the geostatic stress analysis step, the initial stress field is generated by the self-weight of the soil and the pre-load 50 kPa imposed on the surface of the soil, which is then removed completely with 2 equal time increment steps to generate the over-consolidation of the soil layer in the unloading analysis step. Finally, in the load analysis step, the vertical displacement $U_3 = -0.16 \text{ m}$ is imposed on the footing with 10 equal time increment steps. The displacement field at the end of the loading process is presented in Fig. 10(b). In Fig. 11, identical load–displacement responses and the stress paths are noted for the ABAQUS/Standard default algorithm and the proposed algorithm, which further proves the reasonability of the proposed algorithm and the correctness of UMAT. In addition, the shrinkage of the yield surface indicates that the soil near the integral point has softened obviously, which also validates the effectiveness of the proposed algorithm in the softening stage. On the other hand, the agreement between the results of the CTO and the CON shows that the control equations of the global problem guarantee that the search behaviour based on these two tangent operators will reach the same optimal point

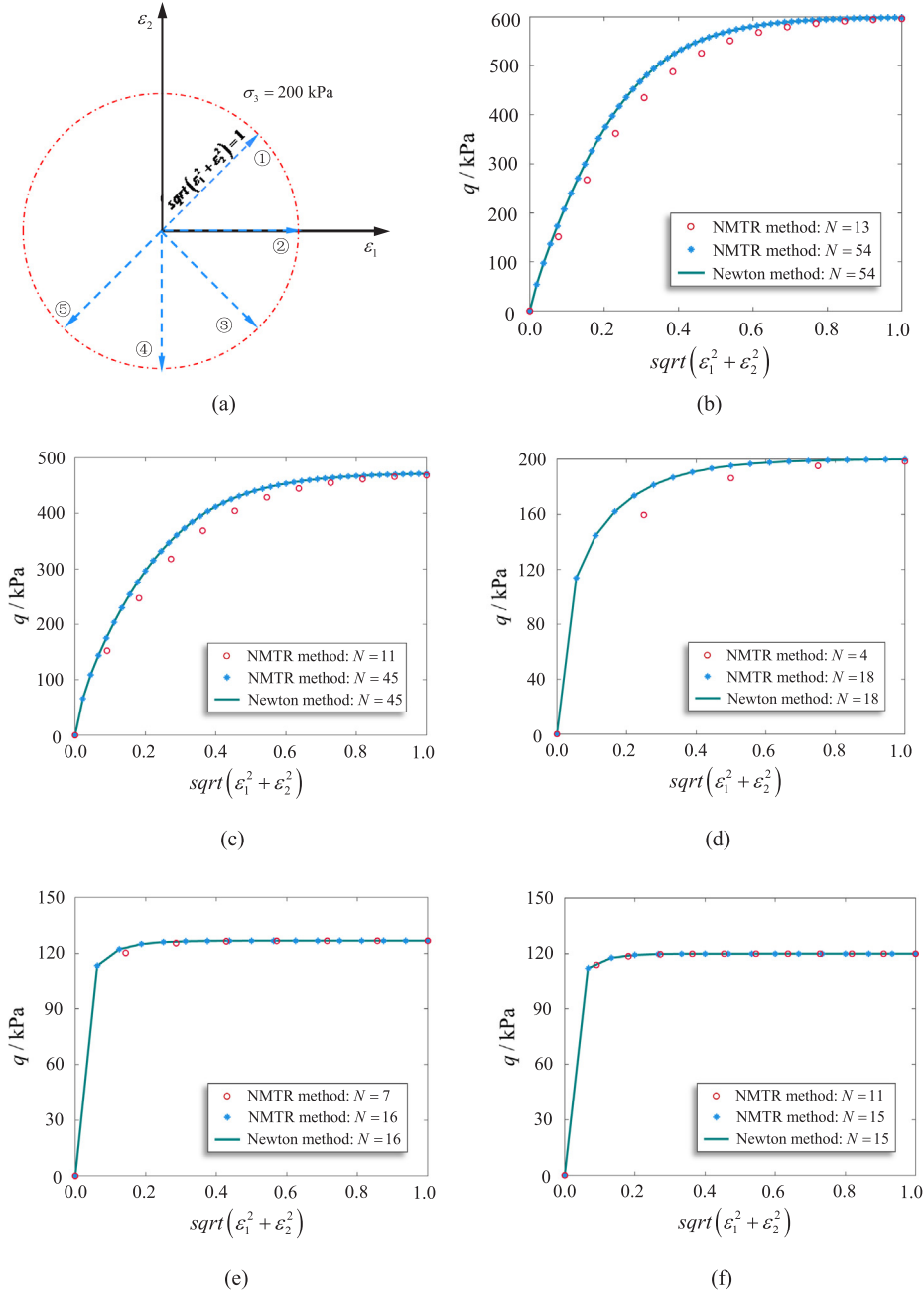


Fig. 8. Convergence test with a single element: (a) considered loading paths; (b) result in of path 1; (c) result of path 2; (d) result of path 3; (e) result of path 4; (f) result of in path 5.

after iterative convergence; namely, the use of the CON does not affect the calculation results on the premise of convergence. However, it must be pointed out that the selections of the two tangent operators lead to differences in computational efficiency.

Fig. 12 shows the convergence process of the normalized largest residual force at the critical node. It is observed that the convergence speed of the NMTR method with the CON is much slower than the ABAQUS/Standard default algorithm and the NMTR method with the CTO. Compared with the examples using CTO, more than three

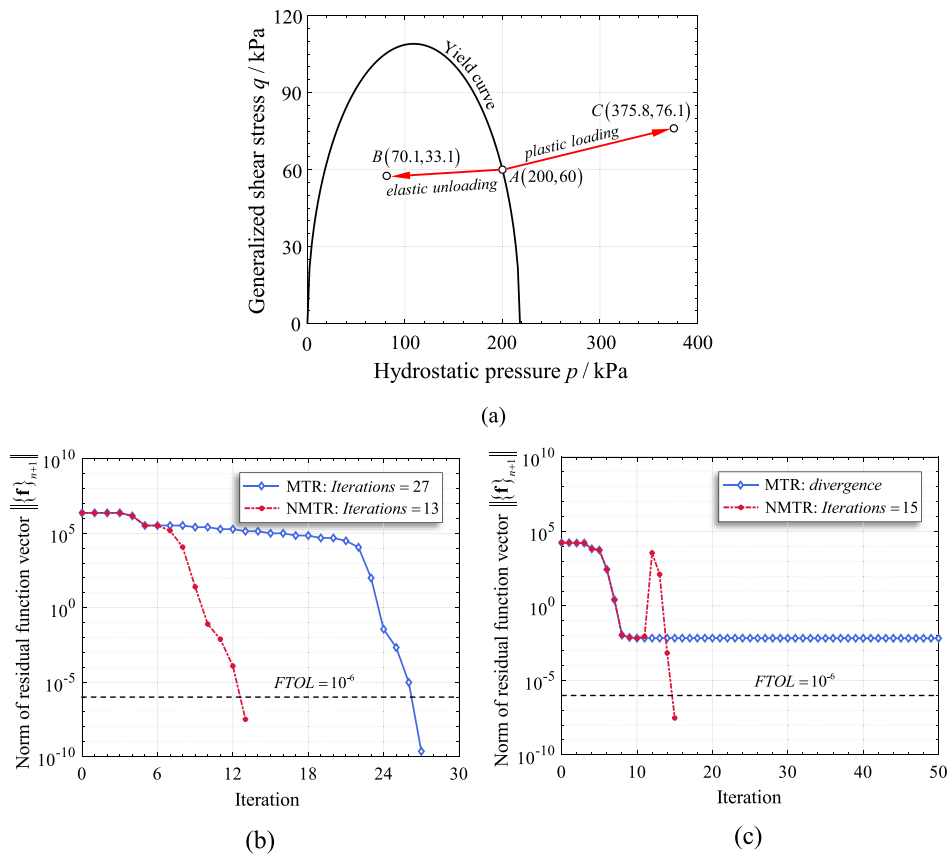


Fig. 9. Comparison between the non-monotonic strategy and monotonic strategy: (a) load paths; (b) loading case; (c) unloading case.

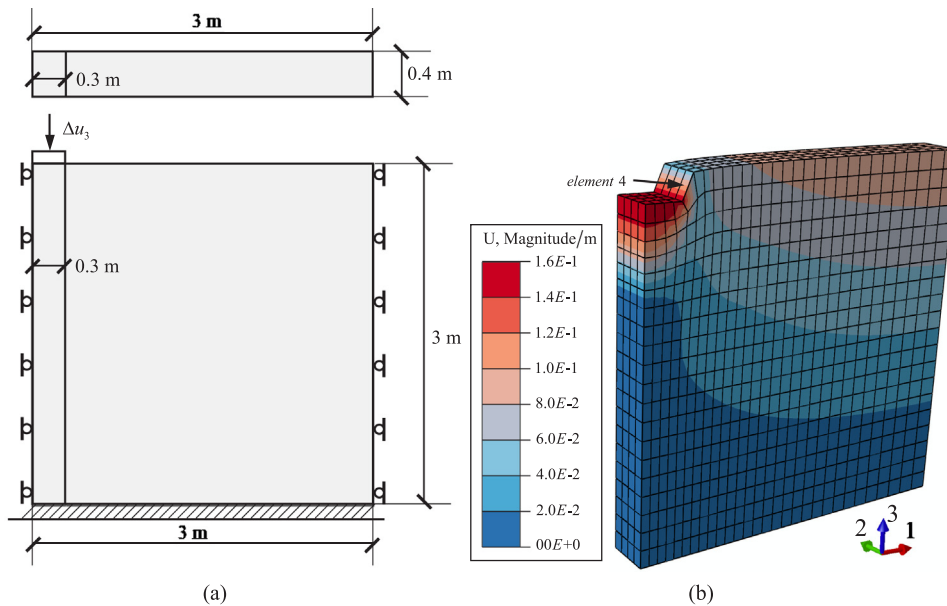


Fig. 10. Collapse of a rigid strip footing: (a) model geometry; (b) finite element mesh.

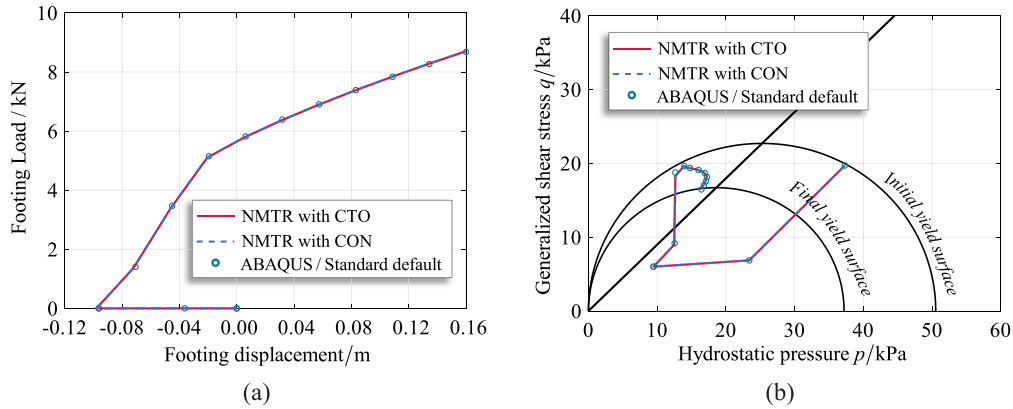


Fig. 11. Comparison between the different algorithm for the strip footing example: (a) footing load versus footing displacement; (b) the stress path of integral point 1 of element 4.

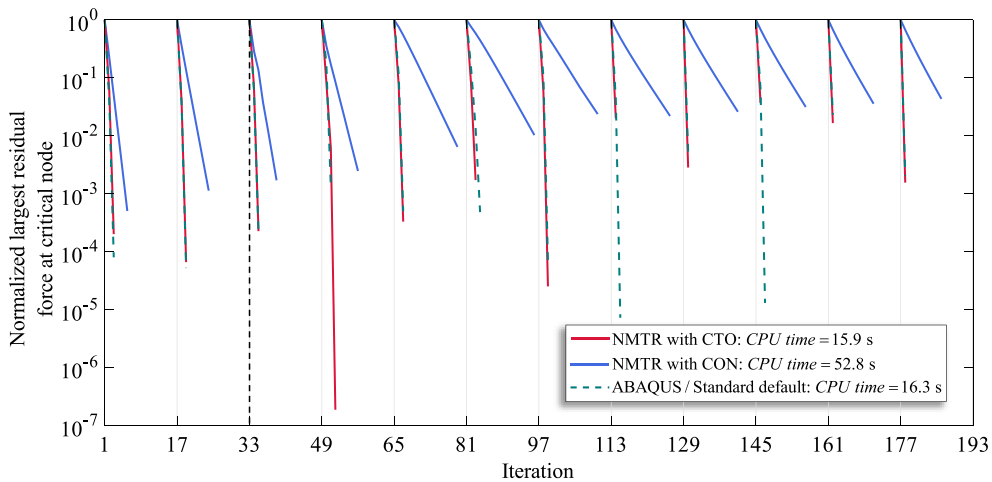


Fig. 12. Change in the normalized largest residual force with the iterations at each time increment step.

times the CPU time is required for the example using the CON. On the other hand, the convergence rates of the ABAQUS/Standard default algorithm and the NMTR method with the CTO are basically the same, which further verifies the effectiveness of smoothing CTO. The number of iterations needed to achieve global equilibrium at each time step is presented in Fig. 13. The comparison result is consistent with the law shown in Fig. 12. Compared with that for the CTO, more than three the number of global equilibrium iterations are required for the CON. Therefore, it is probably best to use the CTO to achieve newton searches of global problems instead of the CON.

6.3. Undrained vertical compressive capacity test of pile foundation

To further assess the algorithm's effectiveness in addressing the geotechnical problems, a vertical compressive capacity test of pile foundation under the undrained condition is simulated. The pile is set in the normally consolidated saturated clay and subjected to vertical displacement load. The pile in the problem is 10 m long with a cross section of diameter 0.5 m. The vertical compressive capacity test of pile foundation can be regarded as an axisymmetric problem. Therefore, the numerical model is simplified as a 2D numerical model shown in Fig. 14 where the geometry information and finite element mesh of the numerical model are presented. The four-node axisymmetric element (CAX4) and four-node axisymmetric pore pressure element (CAX4P) are used for the pile

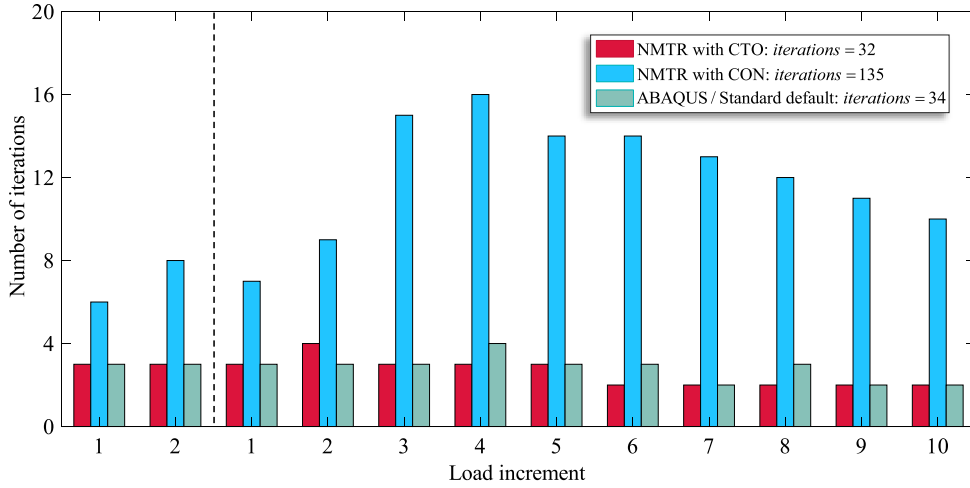


Fig. 13. Number of iterations at each time increment step.

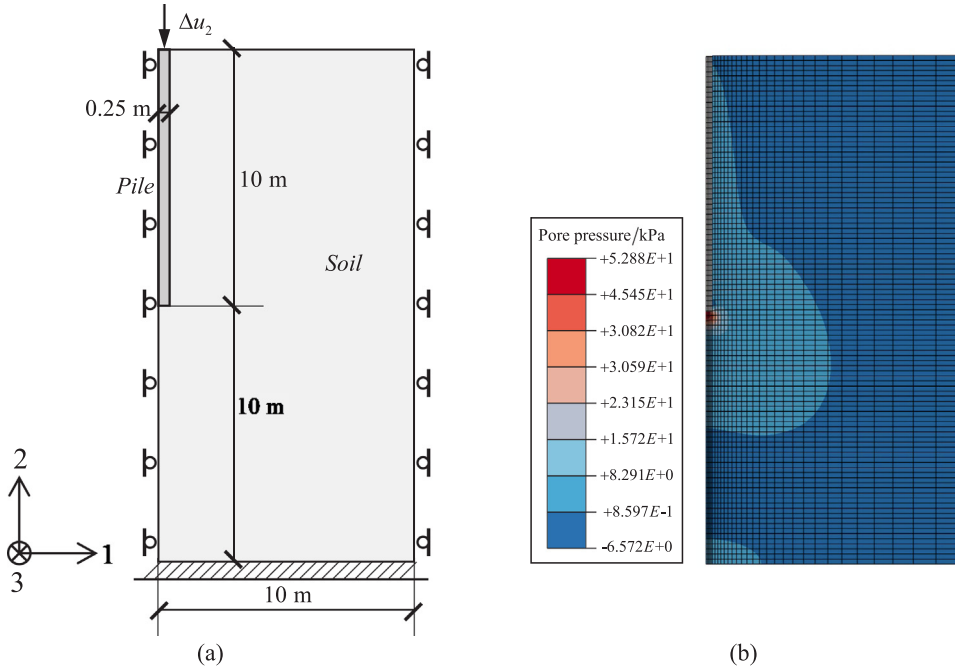


Fig. 14. Vertical compressive capacity test of pile foundation: (a) model geometry; (b) finite element mesh.

and the soil, respectively, to implement such geometric simplification and to consider the pore pressure development. The contact between the pile and the soil is considered as the frictional contact. The friction coefficient is 0.577.

The MCC elastoplastic model and linear elastic model are employed for the soil and the pile, respectively. The material parameters of the pile are set to: $E = 20$ GPa and $\nu = 0.2$. The parameters of the MCC model are set to: $M = 1.2$, $\lambda = 0.2$, $\kappa = 0.04$, $\nu = 0.35$, $e_1 = 2$, and $k_p = 3.6 \times 10^{-4}$ m/h where k_p is the permeability [65]. The submerged weight of the soil and the weight of the water are set to 8 kN/m^3 and 10 kN/m^3 , respectively. In the numerical simulation, the initial stress field is generated in the geostatic stress analysis step where the except that the pore water pressure boundary condition on the top of the soil is set to $u_p = 0$ kPa, the other boundaries are

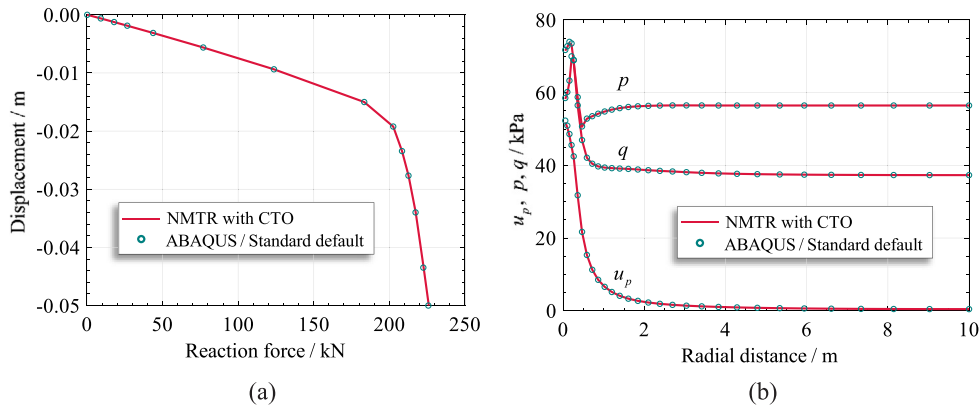


Fig. 15. Comparison between the different algorithms for the pile foundation example: (a) reaction force versus displacement; (b) distributions of pore pressure, hydrostatic pressure, and generalized shear stress along the radial direction.

set to undrained. The vertical displacement load is applied on the top of the pile in the loading analysis step of the soils type. The total time is set to 1 h to approximate undrained loading condition. The initial time increment and the maximum time increment are set to 0.05 h and 0.5 h, respectively. The pore pressure field at the end of loading is shown in Fig. 14(b). The identical reaction force and displacement curves of the top of the pile are presented in Fig. 15(a) for the two algorithms. Fig. 15(b) shows the change of pore pressure and the stress along the radial direction at ten metres depth. It can also be found that the results of the proposed algorithm are in good agreement with that of the ABAQUS/Standard default algorithm.

6.4. Cylindrical sample with cyclically combined tension and shear

Finally, the improvement of the proposed algorithm on the convergence behaviour of the global solution is further investigated. Here, a problem involving cylinder sample with combined tension and shear is considered; this problem has been used to evaluate the performance of the MTR method in Shterenlikht and Alexander's [33] and Lester and Scherzinger's [30] works. One major difference between the question in this section and the one in the literature [30,33] is that cyclic loading with more demand is used instead of monotonic loading to obtain more complex stress conditions. In addition, the ABAQUS/Standard default stress updating algorithm is used as a comparison object.

The cylinder in the problem is 0.2 m long with a cross section of diameter 0.1 m, as shown in Fig. 16. The initial stress state, the pre-consolidation pressure, and material parameters of the first example in Section 6.1 are also used here. The origin of the global coordinate system is at the centre of the bottom face of the cylinder. All degrees of freedom of the bottom face of the cylinder are fixed. The top face of the cylinder is fully coupled to a reference point (0, 0, 0.25). The displacement load in the 1-direction is imposed on the top face of the cylinder by the reference point. In this example, two load cases and three discretization types are used to evaluate the performance of the algorithm. Fig. 17(a) shows the displacement time-history curve in the loading case *a*, which is applied in a single load analysis step with 40 equidistant time steps. For the example analysis in the loading case *a*, only the discretization *a* is used. The loading condition repeatedly suffering the stress behaviour transition from elasticity to plasticity can be realized by the gradually increasing cyclic load in Fig. 17(a). The time-history curves of the reaction force of the reference point for the two algorithms are demonstrated in Fig. 17(b). The results obtained from the two algorithms are in good agreement, which also shows that the smoothing function is effective in replacing the *KT* conditions.

The numbers of global equilibrium iterations at each time step for the two algorithms are depicted in Fig. 18. An oscillating number of iterations is observed for the ABAQUS/Standard default algorithm, while almost all global equilibrium iterations can converge in two steps for the proposed algorithm. This trend may indicate that the smoothing CTO seems to perform better than the CTO that follows the operator splitting technique under

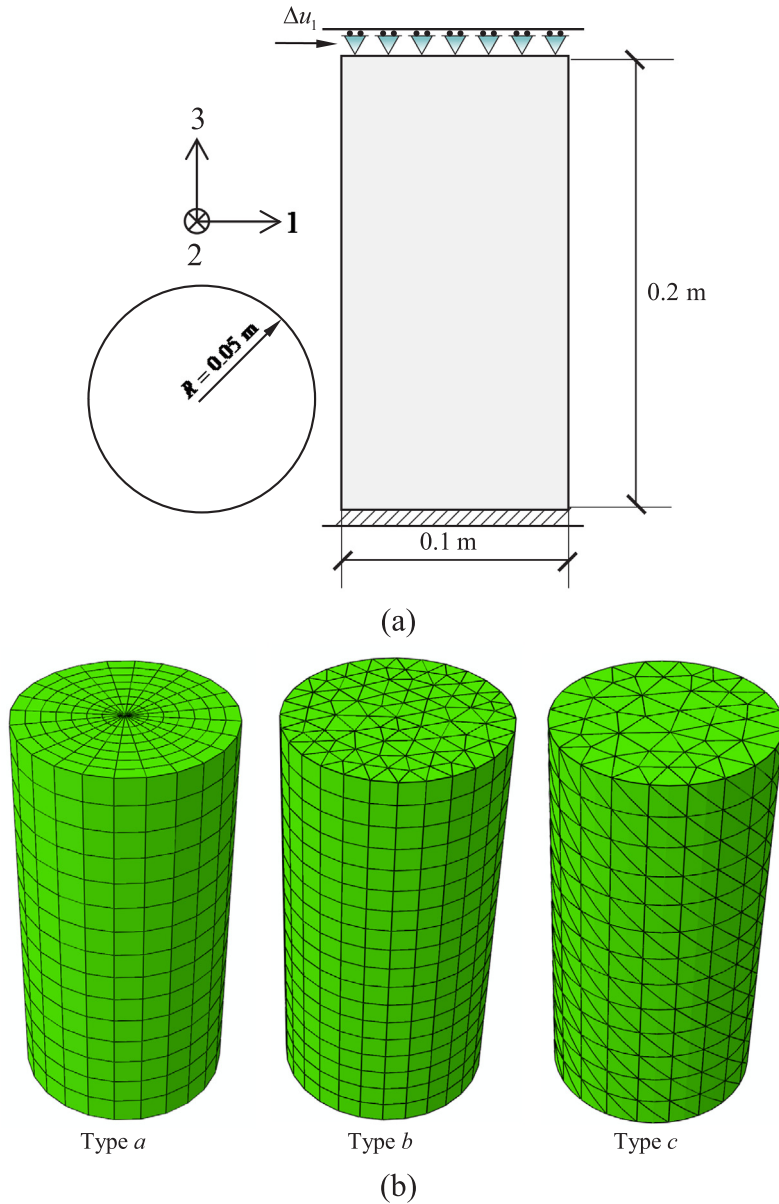


Fig. 16. Geometry and discretization of the cylinder sample: (a) geometry; (b) discretization.

the condition of repeated loading/unloading because the smooth function eliminates the non-smoothness of the elastoplastic problem.

Furthermore, to investigate the influence of the NMTR method on the global calculation, the displacement time history presented in Fig. 19(a) is imposed on the top face of the cylinder by the reference point. During the calculation process, a total of 10 load analysis steps are used, where the new analysis step is activated once the direction of the displacement load changes. During each analysis step, the initial time step is set to 0.1 s. The adjustment of the following time step size is controlled by ABAQUS default *automatic time stepping*, and the maximum time step size is the total time of the current load analysis step. In addition, all three discretization types in Fig. 16(b) are used to further assess the algorithm performance.

The reaction displacement curves of the reference point in the 1-direction are demonstrated for the different discretizations types in Fig. 19(b) to (d). The Job information with the different discretizations is summarized in

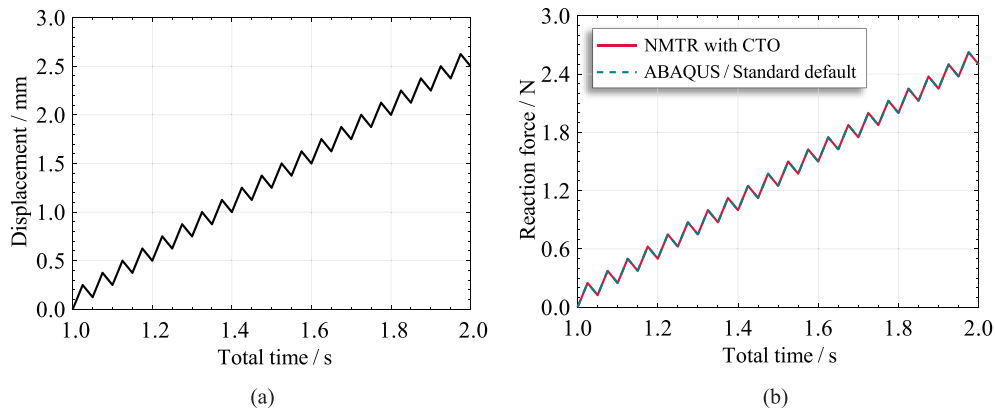


Fig. 17. Load and response in case *a*: (a) displacement load; (b) reaction force response in the 1-direction.

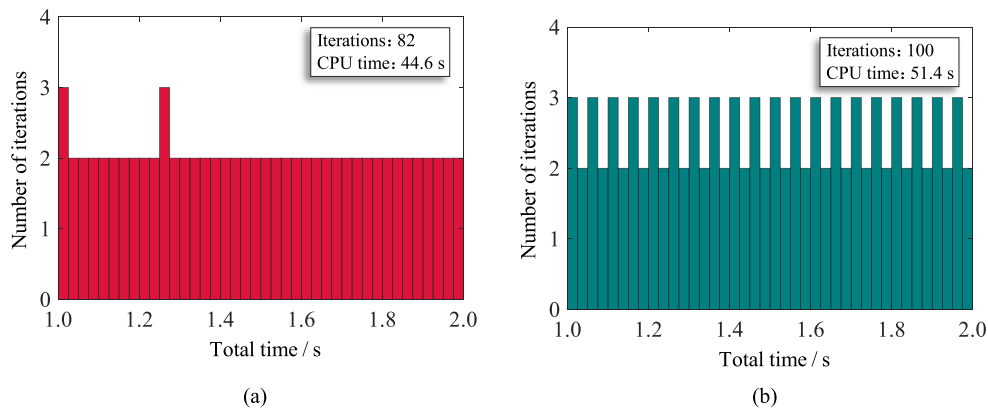


Fig. 18. Global equilibrium iterations at each time step: (a) NMTR with the smoothing CTO; (b) ABAQUS/Standard default integration algorithm.

Table 1

Job information with the different discretizations.

Discretization	Algorithms	Job status (Total time/s)	CPU time/s	Elements number	Number of total/failed load increments		Number of total/failed iterations	
<i>a</i>	ABAQUS/Standard	Completed (51)	428.2	2520	587	118	793	140
	NMTR with CTO		192.4		150	14	376	21
<i>b</i>	ABAQUS/Standard	Completed (51)	315.9	2628	1102	240	1370	254
	NMTR with CTO		98.3		170	19	467	31
<i>c</i>	ABAQUS/Standard	Aborted (10.2964)	376.9	2771	404	97	588	151
	NMTR with CTO	Aborted (12.1412/	227.8		140	30	356	97
		10.2964)	(133.7)		(84)	(13)	(210)	(26)

Table 1. It should be noted that, in the case of the discretization *c*, the ABAQUS/Standard default algorithm is aborted earlier than the proposed algorithm. Therefore, Table 1 additionally supplements the Job information of the proposed algorithm under the same analysis step time as the ABAQUS/Standard default algorithm so as to compare the computational efficiency difference between them under the same condition. It can be found that the reaction displacement curves obtained by the proposed algorithm are in good agreement with those obtained by

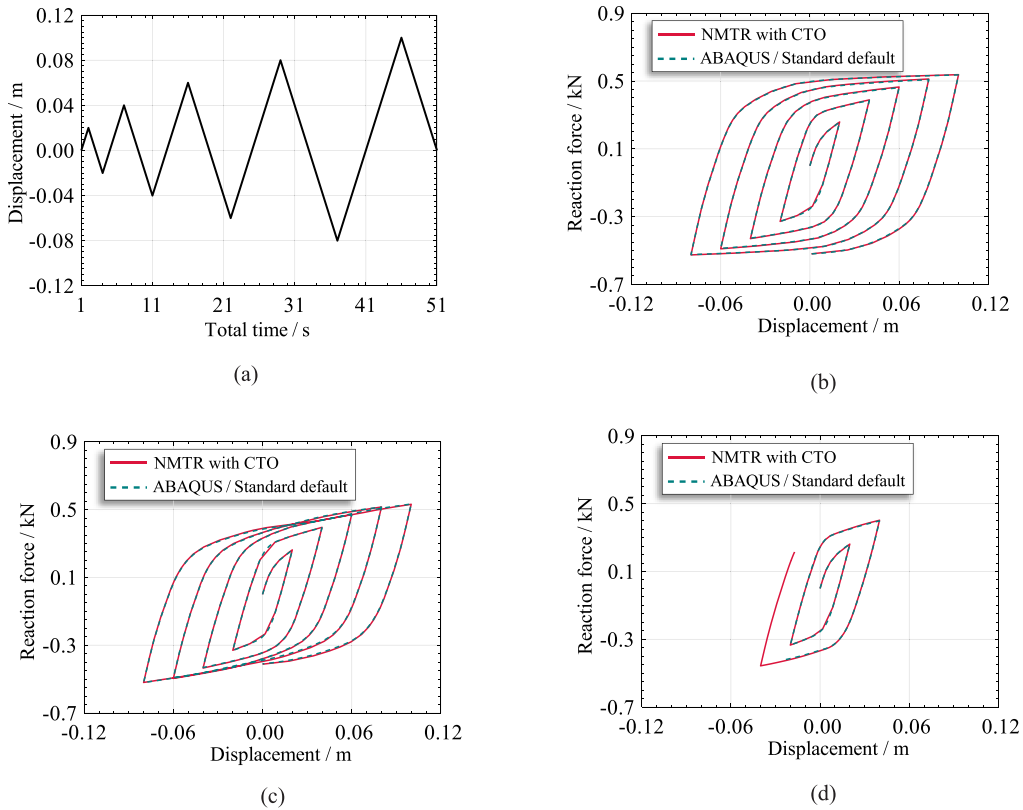


Fig. 19. Load and response in case *b*: (a) displacement load; (b) reaction force versus displacement for discretization *a*; (c) reaction force versus displacement for discretization *b*; (d) reaction force versus displacement for discretization *c*.

the ABAQUS algorithm, while the former has significant computational savings compared with the latter. In what follows, the result with the discretization *a* is used as an example to present the convergence advantages of the proposed algorithm in more detail. Fig. 20(a) shows that the changes in the time step size for the two algorithms for the discretization *a*. During the initial loading period, the time step sizes of the two algorithms are basically the same. Then, compared with those of the ABAQUS/Standard default algorithm, larger time steps are allowed for the proposed stress updating algorithm. The CPU time consumed by the proposed algorithm is only 44.9% of that of the ABAQUS/Standard default algorithm. There are a few main reasons for this difference. On the one hand, in contrast to the ABAQUS/Standard default algorithm, a local problem may be solved by the NMTR method even under larger, more complex loadings, thereby enabling the global solution to be found. On the other hand, as discussed earlier, the use of the smoothing CTO is capable of effectively guaranteeing the convergence behaviour of the global solution even under repeated loading and unloading conditions because the smoothing function eliminates the non-smoothness of the elastoplastic problem. Based on these two improvements, costly cutbacks to the time step size are avoided due to the effective treatment of the non-smoothness and nonlinearity of the initial value problem by the proposed algorithm. As shown in Fig. 20(b) and (c), the number of failed attempts encountered by the proposed algorithm is only 11.9% of that of the ABAQUS/Standard default algorithm, and for the number of global equilibrium iterations wasted on the failed attempts, the former is only 15.0% of the latter. Therefore, during each load analysis step, the displacement load can be applied in fewer incremental steps by the proposed algorithm, as shown in Fig. 20(d). During the whole loading process, the number of total load increment steps required by the proposed algorithm is only 25.6% of that required by the ABAQUS/Standard default algorithm. Correspondingly, the number of global equilibrium iterations occupying the main computing resources can be reduced by the proposed algorithm, as shown in Fig. 20(e). The number of total global equilibrium iterations required by the proposed algorithm is only 47.4% of that required by the ABAQUS/Standard default algorithm.

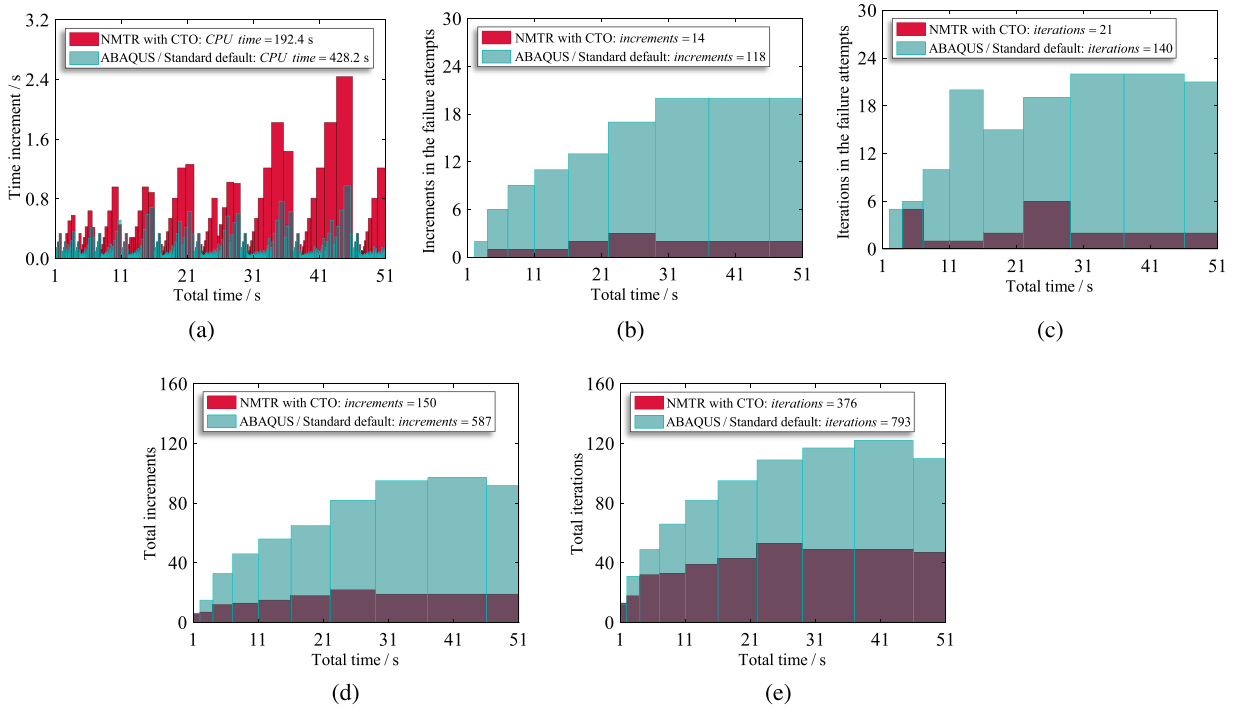


Fig. 20. Convergence behaviour at each load analysis step: (a) change in the time step size; (b) the number of load increments in the failed attempts; (c) the number of global equilibrium iterations in the failed attempts; (d) the number of total load increments; (e) the number of total global equilibrium iterations.

7. Conclusion

A stress updating algorithm for evaluating internal state variables and the CTO for forming global stiffness play a striking role in nonlinear finite-element analysis involving a constitutive model. In this paper, an unconstrained stress updating strategy is proposed based on the *FB* smoothing function. The smoothing CTO and CON corresponding to this strategy are derived. In particular, the algebraic and geometric interpretations of the two tangent operators are discussed and analysed in depth. Furthermore, the NMTR method with strong convergence and robustness is used to solve the nonlinear governing equations of the MCC model. Finally, the MCC model is implemented in ABAQUS via UMAT based on the proposed algorithm to analyse a series of boundary value problems. The correctness and computational efficiency of the proposed algorithm are proven and evaluated via a comparison with the ABAQUS/Standard default stress updating algorithm.

From the perspective of algorithmic structure, there is no need for the loading/unloading judgment step, and the additional consideration of the stress behaviour transition from elasticity to plasticity for the unconstrained stress updating strategy, where the only adjustment made is to use the *FB* smoothing function equivalently instead of *KT* conditions. The elastic CTO and the plastic CTO within the computational framework of the operator splitting technique can be united by the smoothing CTO derived by the unconstrained stress updating strategy. Therefore, the stress update process of the constitutive model can be realized in a simpler procedure. From the perspective of computational efficiency and robustness, larger strain increments are allowed for stress updating in local calculations due to the use of the NMTR method. The non-smoothness in the elastic to plastic transition is also eliminated by the smoothing function. This makes the computational efficiency and robustness of the proposed algorithm under the condition of large time increments and cyclic loadings better than those of the ABAQUS/Standard default. The representative numerical examples show that the CPU time consumed by the proposed algorithm is less than half of that consumed by the ABAQUS/Standard default.

The proposed algorithm presents an exciting possibility for the improvement of the stress updating procedure used for the elastoplastic model. The ability of this algorithm to further address the implementation of more

challenging models (e.g., a multi-yield surface or coupled physics) remains to be explored, whereas it is certain that the result is undoubtedly very promising because the proposed algorithm catches hold of the key to solve initial value problems, i.e., non-smoothness and nonlinearity. The source code provided is capable of creating an easier research environment for these promising works.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Elements in a Jacobian matrix

To facilitate the derivation of the Jacobian matrix $\{\nabla \mathbf{f}\}$ of residual equations in Eq. (34), the derivatives of $\Delta \varepsilon_v^e$ with respect to the unknown variables p_{n+1} , q_{n+1} , $p_{c,n+1}$, and $\Delta \phi_{n+1}$ are derived first. $\Delta \varepsilon_v^e$ can be expressed as follows:

$$\Delta \varepsilon_{v,n+1}^e = \Delta \varepsilon_{v,n+1} - \Delta \varepsilon_{v,n+1}^p = \Delta \varepsilon_{v,n+1} - \Delta \phi_{n+1} (2p_{n+1} - p_{c,n+1}) \quad (\text{A.1})$$

Taking the derivative of Eq. (A.1), we can obtain:

$$\begin{cases} \frac{\partial \Delta \varepsilon_{v,n+1}^p}{\partial p_{n+1}} = 2\Delta \phi_{n+1} \\ \frac{\partial \Delta \varepsilon_{v,n+1}^p}{\partial q_{n+1}} = 0 \\ \frac{\partial \Delta \varepsilon_{v,n+1}^p}{\partial p_{c,n+1}} = -\Delta \phi_{n+1} \\ \frac{\partial \Delta \varepsilon_{v,n+1}^p}{\partial \Delta \phi_{n+1}} = 2p_{n+1} - p_{c,n+1} \end{cases} \quad (\text{A.2})$$

Then, the derivatives of the secant shear modulus $\bar{G}$ with respect to the unknown variables p_{n+1} , q_{n+1} , $p_{c,n+1}$, and $\Delta \phi_{n+1}$ can be easily obtained. The expression of $\bar{G}$ is:

$$\bar{G} = \frac{p_n}{\Delta \varepsilon_{v,n+1}^e} \exp \left[\left(\frac{1+e}{\kappa} \Delta \varepsilon_{v,n+1}^e \right) - 1 \right] r \quad (\text{A.3})$$

Taking the derivatives of Eq. (A.3) and based on the chain rule, we can obtain:

$$\begin{cases} \frac{\partial \bar{G}}{\partial p_{n+1}} = -2 \frac{K_{n+1} - \bar{K}}{\Delta \varepsilon_{v,n+1} - \Delta \varepsilon_{v,n+1}^p} r \Delta \phi_{n+1} \\ \frac{\partial \bar{G}}{\partial q_{n+1}} = 0 \\ \frac{\partial \bar{G}}{\partial p_{c,n+1}} = \frac{K_{n+1}^k - \bar{K}}{\Delta \varepsilon_{v,n+1}^e} r \Delta \phi_{n+1} \\ \frac{\partial \bar{G}}{\partial \Delta \phi_{n+1}} = -\frac{K_{n+1}^k - \bar{K}}{\Delta \varepsilon_{v,n+1}^e} r (2p_{n+1} - p_{c,n+1}) \end{cases} \quad (\text{A.4})$$

In what follows, the elements in the Jacobian matrix of the residual equation are given successively. The derivatives of f_1 are

$$\begin{cases} f_{1,1} = \frac{\partial f_1}{\partial p_{n+1}} = c_d (1 + 2\Delta\phi_{n+1} K_{n+1}) \\ f_{1,2} = \frac{\partial f_1}{\partial q_{n+1}} = 0 \\ f_{1,3} = \frac{\partial f_1}{\partial p_{c,n+1}} = -c_d \Delta\phi_{n+1} K_{n+1} \\ f_{1,4} = \frac{\partial f_1}{\partial \Delta\phi_{n+1}} = c_d (2p_{n+1} - p_{c,n+1}) K_{n+1} \end{cases} \quad (\text{A.5})$$

where

$$K_{n+1} = c_\kappa p_n \exp \{c_\kappa [\Delta\varepsilon_{v,n+1} - \Delta\phi_{n+1} (2p_{n+1} - p_{c,n+1})]\} \quad (\text{A.6})$$

The derivatives of f_2 are

$$\begin{cases} f_{2,1} = \frac{\partial f_2}{\partial p_{n+1}} = -\sqrt{\frac{3}{2}} c_d \eta \frac{\partial \bar{G}}{\partial p} \left(2\hat{\mathbf{n}} : \Delta\boldsymbol{\gamma}_{n+1} - \frac{6\eta \Delta\phi_{n+1}}{M^2} \|\mathbf{s}_n + 2\bar{G} \Delta\boldsymbol{\gamma}_{n+1}\| \right) \\ f_{2,2} = \frac{\partial f_2}{\partial q_{n+1}} = c_d \\ f_{2,3} = \frac{\partial f_2}{\partial p_{c,n+1}} = -\sqrt{\frac{3}{2}} c_d \eta \frac{\partial \bar{G}}{\partial p_{c,n+1}} \left(2\hat{\mathbf{n}} : \Delta\boldsymbol{\gamma}_{n+1} - \frac{6\eta \Delta\phi_{n+1}}{M^2} \|\mathbf{s}_n + 2\bar{G} \Delta\boldsymbol{\gamma}_{n+1}\| \right) \\ f_{2,4} = \frac{\partial f_2}{\partial \Delta\phi_{n+1}} = -\sqrt{\frac{3}{2}} c_d \eta \left[2\hat{\mathbf{n}} : \Delta\boldsymbol{\gamma}_{n+1} \frac{\partial \bar{G}}{\partial \Delta\phi_{n+1}} - \frac{6\eta}{M^2} \|\mathbf{s}_n + 2\bar{G} \Delta\boldsymbol{\gamma}_{n+1}\| \left(\Delta\phi_{n+1} \frac{\partial \bar{G}}{\partial \Delta\phi_{n+1}} + \bar{G} \right) \right] \end{cases} \quad (\text{A.7})$$

where $\|\cdot\|$ is the 2-norm, $\hat{\mathbf{n}} : \Delta\boldsymbol{\gamma}_{n+1} = \hat{n}_{ij} \gamma_{ij} = n_{11} \gamma_{11} + n_{22} \gamma_{22} + n_{33} \gamma_{33} + 2(n_{12} \gamma_{12} + n_{23} \gamma_{23} + n_{13} \gamma_{13})$, and

$$\begin{cases} \eta = \frac{1}{1 + 6\bar{G} \Delta\phi_{n+1} / M^2} \\ \hat{\mathbf{n}} = \frac{\mathbf{s}_{n+1}}{\|\mathbf{s}_{n+1}\|} \end{cases} \quad (\text{A.8})$$

The derivatives of f_3 are

$$\begin{cases} f_{3,1} = \frac{\partial f_3}{\partial p_{n+1}} = c_d \{-2c_p \Delta\phi_{n+1} p_{c,n} \exp [c_p \Delta\phi_{n+1} (2p_{n+1} - p_{c,n+1})]\} \\ f_{3,2} = \frac{\partial f_3}{\partial q_{n+1}} = 0 \\ f_{3,3} = \frac{\partial f_3}{\partial p_{c,n+1}} = c_d \{1 + c_p \Delta\phi_{n+1} p_{c,n} \exp [c_p \Delta\phi_{n+1} (2p_{n+1} - p_{c,n+1})]\} \\ f_{3,4} = \frac{\partial f_3}{\partial \Delta\phi_{n+1}} = -c_d c_p (2p_{n+1} - p_{c,n+1}) p_{c,n} \exp [c_p \Delta\phi_{n+1} (2p_{n+1} - p_{c,n+1})] \end{cases} \quad (\text{A.9})$$

The derivatives of f_4 are

$$\begin{cases} f_{4,1} = \frac{\partial f_4}{\partial p_{n+1}} = \chi_0 (2p_{n+1} - p_{c,n+1}) \\ f_{4,2} = \frac{\partial f_4}{\partial q_{n+1}} = \chi_0 \frac{2q_{n+1}}{M^2} \\ f_{4,3} = \frac{\partial f_4}{\partial p_{c,n+1}} = -\chi_0 p_{n+1} \\ f_{4,4} = \frac{\partial f_4}{\partial \Delta\phi_{n+1}} = \chi_1 \end{cases} \quad (\text{A.10})$$

where

$$\begin{cases} \chi_0 = \frac{\partial f_4}{\partial f_{n+1}} = \frac{f_{n+1}}{\sqrt{(c_d \Delta \phi_{n+1})^2 + f_{n+1}^2 + 2\beta}} + 1 \\ \chi_1 = \frac{\partial f_4}{\partial \Delta \phi_{n+1}} = \frac{c_d^2 \Delta \phi_{n+1}}{\sqrt{(c_d \Delta \phi_{n+1})^2 + f_{n+1}^2 + 2\beta}} - c_d \end{cases} \quad (\text{A.11})$$

Appendix B. Smoothing consistent tangent operator

Taking the derivative of Eq. (A.3), we can obtain:

$$\frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = \mathbf{1} \otimes \frac{\partial p_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} + \frac{\partial \mathbf{s}_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \quad (\text{B.1})$$

where $\partial p_{n+1} / \partial \boldsymbol{\epsilon}_{n+1}$ is:

$$\frac{\partial p_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = K_{n+1} \left[\mathbf{1} - \Delta \phi_{n+1} \left(2 \frac{\partial p_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} - \frac{\partial p_{c,n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \right) - (2p_{n+1} - p_{c,n+1}) \frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \right] \quad (\text{B.2})$$

where

$$\frac{\partial p_{c,n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = p_{c,n+1} c_p \left[\Delta \phi_{n+1} \left(2 \frac{\partial p_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} - \frac{\partial p_{c,n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \right) + (2p_{n+1} - p_{c,n+1}) \frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \right] \quad (\text{B.3})$$

By simultaneously addressing Eqs. (B.2) and (B.3), the following expressions can be obtained with the only $\partial \Delta \phi_{n+1} / \partial \boldsymbol{\epsilon}_{n+1}$ unknown as follows:

$$\frac{\partial p_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = a_1 K_{n+1} \mathbf{1} + a_2 K_{n+1} \frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \quad (\text{B.4})$$

$$\frac{\partial p_{c,n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = a_3 K_{n+1} \mathbf{1} + a_4 K_{n+1} \frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \quad (\text{B.5})$$

where the coefficients a_1 , a_2 , a_3 , and a_4 are expressed by:

$$\begin{cases} a_0 = 1 + p_{c,n+1} c_p \Delta \phi_{n+1} + 2 \Delta \phi_{n+1} K_{n+1} \\ a_1 = (1 + p_{c,n+1} c_p \Delta \phi_{n+1}) / a_0 \\ a_2 = -(2p_{n+1} - p_{c,n+1}) / a_0 \\ a_3 = 2p_{c,n+1} c_p \Delta \phi_{n+1} / a_0 \\ a_4 = p_{c,n+1} c_p (2p_{n+1} - p_{c,n+1}) / (a_0 K_{n+1}) \end{cases} \quad (\text{B.6})$$

$\partial \Delta \phi_{n+1} / \partial \boldsymbol{\epsilon}_{n+1}$ can be derived by imposing the consistency condition of Eq. (34)₄. Taking the differentiation of Eq. (34)₄ can yield:

$$\frac{\partial f_4}{\partial f} \left(\frac{\partial f}{\partial \mathbf{s}_{n+1}} : \frac{\partial \mathbf{s}_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} + \frac{\partial f}{\partial p_{n+1}} \frac{\partial p_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} + \frac{\partial f}{\partial p_{c,n+1}} \frac{\partial p_{c,n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \right) + \frac{\partial f_4}{\partial \Delta \phi_{n+1}} \frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = 0 \quad (\text{B.7})$$

where the partial derivatives of the yield function with respect to $\mathbf{s}_{n+1}$, p_{n+1} , and $p_{c,n+1}$ are:

$$\begin{cases} \frac{\partial f}{\partial \mathbf{s}_{n+1}} = \frac{3}{M^2} \mathbf{s}_{n+1} \\ \frac{\partial f}{\partial p_{n+1}} = 2p_{n+1} - p_{c,n+1} \\ \frac{\partial f}{\partial p_{c,n+1}} = -p_{n+1} \end{cases} \quad (\text{B.8})$$

Taking the derivative of Eq. (28) with respect to $\boldsymbol{\epsilon}_{n+1}$ and noticing that $\Delta \phi_{n+1}$ and $\bar{G}$ are also functions of $\boldsymbol{\epsilon}_{n+1}$ can yield:

$$\frac{\partial \mathbf{s}_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = 2\eta \left(\Delta \boldsymbol{\gamma}_{n+1} \otimes \frac{\partial \bar{G}}{\partial \boldsymbol{\epsilon}_{n+1}} + \bar{G} \mathbf{P} \right) - \eta^2 (\mathbf{s}_n + 2\bar{G} \Delta \boldsymbol{\gamma}_{n+1}) \otimes \left(\frac{6\bar{G}}{M^2} \frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} + \frac{6\Delta \phi_{n+1}}{M^2} \frac{\partial \bar{G}}{\partial \boldsymbol{\epsilon}_{n+1}} \right) \quad (\text{B.9})$$

The expression of $\partial \bar{G}/\partial \boldsymbol{\epsilon}_{n+1}$ is as follows:

$$\frac{\partial \bar{G}}{\partial \boldsymbol{\epsilon}_{n+1}} = \frac{\partial \bar{G}}{\partial p_{n+1}} \frac{\partial p_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} + \frac{\partial \bar{G}}{\partial \Delta \varepsilon_{v,n+1}^e} \frac{\partial \Delta \varepsilon_{v,n+1}^e}{\partial \boldsymbol{\epsilon}_{n+1}} = \frac{(rK_{n+1} - \bar{G})}{\Delta \varepsilon_{v,n+1}^e} \left(a_1 \mathbf{1} + a_2 \frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \right) \quad (\text{B.10})$$

Putting Eq. (B.10) into Eq. (B.9) can obtain the new expression of $\partial \mathbf{s}_{n+1}/\partial \boldsymbol{\epsilon}_{n+1}$ with the only unknown $\partial \Delta \phi_{n+1}/\partial \boldsymbol{\epsilon}_{n+1}$ as follows:

$$\begin{aligned} \frac{\partial \mathbf{s}_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = & 2\bar{G}\eta \left[\mathbf{P} + \frac{\Delta \boldsymbol{\gamma}_{n+1}}{\bar{G}} \otimes \frac{rK_{n+1} - \bar{G}}{\Delta \varepsilon_{v,n+1}^e} \left(a_1 \mathbf{1} + a_2 \frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \right) \right] \\ & - \frac{2\sqrt{6}q\bar{G}\eta}{M^2} \hat{\mathbf{n}} \otimes \left[\frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} + \frac{\Delta \phi_{n+1}}{\bar{G}} \frac{rK_{n+1} - \bar{G}}{\Delta \varepsilon_{v,n+1}^e} \left(a_1 \mathbf{1} + a_2 \frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} \right) \right] \end{aligned} \quad (\text{B.11})$$

Then, substituting Eqs. (B.4), (B.5), and (B.11) into Eq. (B.7) and rearranging the expression can yield

$$\frac{\partial \Delta \phi_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = b_1 \mathbf{1} + b_2 \hat{\mathbf{n}} \quad (\text{B.12})$$

where

$$\begin{cases} b_0 = \chi_0 \left\{ \left[\left(\frac{\hat{\mathbf{n}}}{\sqrt{6}} : \Delta \boldsymbol{\gamma}_{n+1} - \frac{q\Delta \phi_{n+1}}{M^2} \right) a_2 \frac{rK_{n+1} - \bar{G}}{\Delta \varepsilon_{v,n+1}^e} - \frac{q\bar{G}}{M^2} \right] \frac{12q\eta}{M^2} \right. \\ \quad \left. + [(2a_2 - a_4)p_{n+1} - a_2 p_{c,n+1}] K_{n+1} \right\} + \chi_1 \\ b_1 = -\chi_0 \left[\left(\frac{\hat{\mathbf{n}}}{\sqrt{6}} : \Delta \boldsymbol{\gamma}_{n+1} - \frac{q\Delta \phi_{n+1}}{M^2} \right) \frac{12qa_1\eta}{M^2} \frac{rK_{n+1} - \bar{G}}{\Delta \varepsilon_{v,n+1}^e} + (2a_1 - a_3)p_{n+1}K_{n+1} - a_1 p_{c,n+1}K_{n+1} \right] / b_0 \\ b_2 = -\chi_0 \frac{2\sqrt{6}q\bar{G}\eta}{M^2} / b_0 \end{cases} \quad (\text{B.13})$$

Putting Eq. (B.12) into Eq. (B.4) and Eq. (B.9) can yield:

$$\frac{\partial p_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = a_1 K_{n+1} \mathbf{1} + a_2 K_{n+1} (b_1 \mathbf{1} + b_2 \hat{\mathbf{n}}) \quad (\text{B.14})$$

$$\begin{aligned} \frac{\partial \mathbf{s}_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = & 2\bar{G}\eta \left\{ \mathbf{P} + \frac{rK_{n+1} - \bar{G}}{\bar{G}\Delta \varepsilon_{v,n+1}^e} [(a_1 + a_2 b_1) \Delta \boldsymbol{\gamma}_{n+1} \otimes \mathbf{1} + a_2 b_2 \Delta \boldsymbol{\gamma}_{n+1} \otimes \hat{\mathbf{n}}] \right\} \\ & - \frac{2\sqrt{6}q\bar{G}\eta}{M^2} \left[\left(b_1 + \frac{(rK_{n+1} - \bar{G})\Delta \phi_{n+1}}{\bar{G}\Delta \varepsilon_{v,n+1}^e} (a_1 + a_2 b_1) \right) \hat{\mathbf{n}} \otimes \mathbf{1} \right. \\ & \left. + \left(b_2 + a_2 b_2 \frac{(rK_{n+1} - \bar{G})\Delta \phi_{n+1}}{\bar{G}\Delta \varepsilon_{v,n+1}^e} \right) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}} \right] \end{aligned} \quad (\text{B.15})$$

Finally, putting Eqs. (B.14) and (B.15) into Eq. (B.1), the CTO can result in the following:

$$\begin{aligned} \frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \boldsymbol{\epsilon}_{n+1}} = & 2\bar{G}\eta \mathbf{P} + (a_1 + a_2 b_1) K_{n+1} \mathbf{1} \otimes \mathbf{1} + a_2 b_2 K_{n+1} \mathbf{1} \otimes \hat{\mathbf{n}} + 2\eta \frac{rK_{n+1} - \bar{G}}{\Delta \varepsilon_{v,n+1}^e} (a_1 + a_2 b_1) \Delta \boldsymbol{\gamma}_{n+1} \otimes \mathbf{1} \\ & + 2a_2 \eta b_2 \frac{rK_{n+1} - \bar{G}}{\Delta \varepsilon_{v,n+1}^e} \Delta \boldsymbol{\gamma}_{n+1} \otimes \hat{\mathbf{n}} - \frac{2\sqrt{6}q\eta}{M^2} \left[b_1 \bar{G} + \frac{(rK_{n+1} - \bar{G})\Delta \phi_{n+1}}{\Delta \varepsilon_{v,n+1}^e} (a_1 + a_2 b_1) \right] \hat{\mathbf{n}} \otimes \mathbf{1} \\ & - \frac{2\sqrt{6}q\eta}{M^2} \left(b_2 \bar{G} + a_2 b_2 \frac{(rK_{n+1} - \bar{G})\Delta \phi_{n+1}}{\Delta \varepsilon_{v,n+1}^e} \right) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}} \end{aligned} \quad (\text{B.16})$$

Based on Eq. (B.16), the coefficients of the tensor in Eq. (70) can be given as follows:

$$\begin{cases} c_0 = 2\bar{G}\eta \\ c_1 = (a_1 + a_2b_1)K_{n+1} \\ c_2 = a_2b_2K_{n+1} \\ c_3 = 2(a_1\eta + a_2\eta b_1) \frac{rK_{n+1} - \bar{G}}{\Delta\epsilon_{v,n+1}^e} \\ c_4 = 2a_2\eta b_2 \frac{rK_{n+1} - \bar{G}}{\Delta\epsilon_{v,n+1}^e} \\ c_5 = -\frac{2\sqrt{6}q\eta}{M^2} \left[b_1\bar{G} + (a_1 + a_2b_1) \frac{(rK_{n+1} - \bar{G}) \Delta\phi_{n+1}}{\Delta\epsilon_{v,n+1}^e} \right] \\ c_6 = -\frac{2\sqrt{6}q\eta}{M^2} \left[b_2\bar{G} + a_2b_2 \frac{(rK_{n+1} - \bar{G}) \Delta\phi_{n+1}}{\Delta\epsilon_{v,n+1}^e} \right] \end{cases} \quad (\text{B.17})$$

Appendix C. Matrix representation of tensors

In the numerical implementation of the model, the second-order tensor and fourth-order tensor can be expanded into a 6-by-1 matrix and a 6-by-6 matrix, respectively, for more efficient computing due to their inherent symmetry.

The matrix expressions of the second order in this paper are given as follows:

$\mathbf{1} = \delta_{ij}\mathbf{e}_i \otimes \mathbf{e}_j$, where $\mathbf{e}_i$ and $\mathbf{e}_j$ are the orthonormal bases of the second-order tensor, and the component δ_{ij} is the Kronecker delta, which can be expressed by:

$$\delta_{ij} = [1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0]^T \quad (\text{C.1})$$

$\partial f / \partial \boldsymbol{\sigma} = \partial f / \partial \sigma_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$, where the components of $\partial f / \partial \sigma_{ij}$ can be expressed by:

$$\frac{\partial f}{\partial \sigma_{ij}} = [\partial f / \partial \sigma_{11} \quad \partial f / \partial \sigma_{22} \quad \partial f / \partial \sigma_{33} \quad 2\partial f / \partial \sigma_{12} \quad 2\partial f / \partial \sigma_{23} \quad 2\partial f / \partial \sigma_{13}]^T \quad (\text{C.2})$$

$\Delta \boldsymbol{\sigma} = \Delta \sigma_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$, where the components of $\Delta \sigma_{ij}$ can be expressed by:

$$\Delta \sigma_{ij} = [\Delta \sigma_{11} \quad \Delta \sigma_{22} \quad \Delta \sigma_{33} \quad \Delta \sigma_{12} \quad \Delta \sigma_{23} \quad \Delta \sigma_{13}]^T \quad (\text{C.3})$$

$\Delta \boldsymbol{\epsilon}_{n+1} = \Delta \epsilon_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$, where the components of $\Delta \epsilon_{ij}$ can be expressed by:

$$\Delta \epsilon_{ij} = [\Delta \epsilon_{11} \quad \Delta \epsilon_{22} \quad \Delta \epsilon_{33} \quad \Delta \epsilon_{12} \quad \Delta \epsilon_{23} \quad \Delta \epsilon_{13}]^T \quad (\text{C.4})$$

$\Delta \boldsymbol{\gamma}_{n+1} = \Delta \gamma_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$, where the components of $\Delta \gamma_{ij} = \Delta \epsilon_{ij} - \delta_{ij} \Delta \epsilon_v / 3$ can be expressed by:

$$\Delta \gamma_{ij} = [\Delta \gamma_{11} \quad \Delta \gamma_{22} \quad \Delta \gamma_{33} \quad \Delta \gamma_{12} \quad \Delta \gamma_{23} \quad \Delta \gamma_{13}]^T \quad (\text{C.5})$$

$\hat{\mathbf{n}} = \hat{n}_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$, where the components of $\hat{n}_{ij}$ can be expressed by:

$$\hat{n}_{ij} = [\hat{n}_{11} \quad \hat{n}_{22} \quad \hat{n}_{33} \quad \hat{n}_{12} \quad \hat{n}_{23} \quad \hat{n}_{13}]^T = \frac{1}{\|\mathbf{s}\|} [s_{11} \quad s_{22} \quad s_{33} \quad s_{12} \quad s_{23} \quad s_{13}]^T \quad (\text{C.6})$$

The matrix expression of the fourth order is:

$\mathbf{1} = \mathbf{1} \otimes \mathbf{1} = \delta_{ij}\delta_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$, where $\mathbf{e}_i$, $\mathbf{e}_j$, $\mathbf{e}_k$, and $\mathbf{e}_l$ are the orthonormal bases of the fourth-order tensor and the components $\delta_{ij}\delta_{kl}$ can be expressed by:

$$\delta_{ij}\delta_{kl} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{C.7})$$

$\mathbf{I}^{\text{vol}} = \frac{1}{3} \mathbf{1} \otimes \mathbf{1} = \frac{1}{3} \delta_{ij} \delta_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$, where the components $\frac{1}{3} \delta_{ij} \delta_{kl}$ can be expressed by:

$$\frac{1}{3} \delta_{ij} \delta_{kl} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{C.8})$$

$\mathbf{I}^{\text{sym}} = \frac{1}{2} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$, where the components $\frac{1}{2} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$ can be expressed by:

$$\frac{1}{2} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 \end{bmatrix} \quad (\text{C.9})$$

$\bar{\mathbf{D}} = 3\mathbf{I}^{\text{vol}} (\bar{K} - \frac{2}{3} \bar{G}) + 2\mathbf{I}^{\text{sym}} \bar{G} = \bar{D}_{ijkl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$, where the components $\bar{D}_{ijkl}$ can be expressed by:

$$\bar{D}_{ijkl} = \delta_{ij} \delta_{kl} \left(\bar{K} - \frac{2}{3} \bar{G} \right) + (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \bar{G} = \begin{bmatrix} \bar{K} + \frac{1}{3} \bar{G} & \bar{K} - \frac{2}{3} \bar{G} & \bar{K} - \frac{2}{3} \bar{G} & 0 & 0 & 0 \\ \bar{K} - \frac{2}{3} \bar{G} & \bar{K} + \frac{1}{3} \bar{G} & \bar{K} - \frac{2}{3} \bar{G} & 0 & 0 & 0 \\ \bar{K} - \frac{2}{3} \bar{G} & \bar{K} - \frac{2}{3} \bar{G} & \bar{K} + \frac{1}{3} \bar{G} & 0 & 0 & 0 \\ 0 & 0 & 0 & \bar{G} & 0 & 0 \\ 0 & 0 & 0 & 0 & \bar{G} & 0 \\ 0 & 0 & 0 & 0 & 0 & \bar{G} \end{bmatrix} \quad (\text{C.10})$$

$\mathbf{P}$ is the fourth-order projection tensor and $\mathbf{P} = (\delta_{ik} \delta_{jl} - \frac{1}{3} \delta_{ij} \delta_{kl}) \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$ where the components $\delta_{ik} \delta_{jl} - \frac{1}{3} \delta_{ij} \delta_{kl}$ can be expressed by:

$$\delta_{ik} \delta_{jl} - \frac{1}{3} \delta_{ij} \delta_{kl} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} & 0 & 0 & 0 \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} & 0 & 0 & 0 \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix} \quad (\text{C.11})$$

$\mathbf{1} \otimes \hat{\mathbf{n}} = \delta_{ij} \hat{n}_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$, where the components $\delta_{ij} \hat{n}_{kl}$ can be expressed by:

$$\delta_{ij} \hat{n}_{kl} = \begin{bmatrix} \hat{n}_{11} & \hat{n}_{22} & \hat{n}_{33} & \hat{n}_{12} & \hat{n}_{23} & \hat{n}_{13} \\ \hat{n}_{11} & \hat{n}_{22} & \hat{n}_{33} & \hat{n}_{12} & \hat{n}_{23} & \hat{n}_{13} \\ \hat{n}_{11} & \hat{n}_{22} & \hat{n}_{33} & \hat{n}_{12} & \hat{n}_{23} & \hat{n}_{13} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{C.12})$$

$\Delta \gamma_{n+1} \otimes \mathbf{1} = \gamma_{ij} \delta_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$, where the components $\gamma_{ij} \delta_{kl}$ can be expressed by:

$$\Delta \gamma_{ij} \delta_{kl} = \begin{bmatrix} \Delta \gamma_{11} & \Delta \gamma_{11} & \Delta \gamma_{11} & 0 & 0 & 0 \\ \Delta \gamma_{22} & \Delta \gamma_{22} & \Delta \gamma_{22} & 0 & 0 & 0 \\ \Delta \gamma_{33} & \Delta \gamma_{33} & \Delta \gamma_{33} & 0 & 0 & 0 \\ \Delta \gamma_{12} & \Delta \gamma_{12} & \Delta \gamma_{12} & 0 & 0 & 0 \\ \Delta \gamma_{23} & \Delta \gamma_{23} & \Delta \gamma_{23} & 0 & 0 & 0 \\ \Delta \gamma_{13} & \Delta \gamma_{13} & \Delta \gamma_{13} & 0 & 0 & 0 \end{bmatrix} \quad (\text{C.13})$$

$\Delta \gamma_{n+1} \otimes \hat{\mathbf{n}} = \Delta \gamma_{ij} \hat{n}_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$, where the components $\Delta \gamma_{ij} \hat{n}_{kl}$ can be expressed by:

$$\Delta \gamma_{ij} \hat{n}_{kl} = \begin{bmatrix} \Delta \gamma_{11} \hat{n}_{11} & \Delta \gamma_{11} \hat{n}_{22} & \Delta \gamma_{11} \hat{n}_{33} & \Delta \gamma_{11} \hat{n}_{12} & \Delta \gamma_{11} \hat{n}_{23} & \Delta \gamma_{11} \hat{n}_{13} \\ \Delta \gamma_{22} \hat{n}_{11} & \Delta \gamma_{22} \hat{n}_{22} & \Delta \gamma_{22} \hat{n}_{33} & \Delta \gamma_{22} \hat{n}_{12} & \Delta \gamma_{22} \hat{n}_{23} & \Delta \gamma_{22} \hat{n}_{13} \\ \Delta \gamma_{33} \hat{n}_{11} & \Delta \gamma_{33} \hat{n}_{22} & \Delta \gamma_{33} \hat{n}_{33} & \Delta \gamma_{33} \hat{n}_{12} & \Delta \gamma_{33} \hat{n}_{23} & \Delta \gamma_{33} \hat{n}_{13} \\ \Delta \gamma_{12} \hat{n}_{11} & \Delta \gamma_{12} \hat{n}_{22} & \Delta \gamma_{12} \hat{n}_{33} & \Delta \gamma_{12} \hat{n}_{12} & \Delta \gamma_{12} \hat{n}_{23} & \Delta \gamma_{12} \hat{n}_{13} \\ \Delta \gamma_{23} \hat{n}_{11} & \Delta \gamma_{23} \hat{n}_{22} & \Delta \gamma_{23} \hat{n}_{33} & \Delta \gamma_{23} \hat{n}_{12} & \Delta \gamma_{23} \hat{n}_{23} & \Delta \gamma_{23} \hat{n}_{13} \\ \Delta \gamma_{13} \hat{n}_{11} & \Delta \gamma_{13} \hat{n}_{22} & \Delta \gamma_{13} \hat{n}_{33} & \Delta \gamma_{13} \hat{n}_{12} & \Delta \gamma_{13} \hat{n}_{23} & \Delta \gamma_{13} \hat{n}_{13} \end{bmatrix} \quad (\text{C.14})$$

$\hat{\mathbf{n}} \otimes \mathbf{1} = \hat{n}_{ij} \delta_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$, where the components $\hat{n}_{ij} \delta_{kl}$ can be expressed by:

$$\hat{n}_{ij} \delta_{kl} = \begin{bmatrix} \hat{n}_{11} & \hat{n}_{11} & \hat{n}_{11} & 0 & 0 & 0 \\ \hat{n}_{22} & \hat{n}_{22} & \hat{n}_{22} & 0 & 0 & 0 \\ \hat{n}_{33} & \hat{n}_{33} & \hat{n}_{33} & 0 & 0 & 0 \\ \hat{n}_{12} & \hat{n}_{12} & \hat{n}_{12} & 0 & 0 & 0 \\ \hat{n}_{23} & \hat{n}_{23} & \hat{n}_{23} & 0 & 0 & 0 \\ \hat{n}_{13} & \hat{n}_{13} & \hat{n}_{13} & 0 & 0 & 0 \end{bmatrix} \quad (\text{C.15})$$

$\hat{\mathbf{n}} \otimes \hat{\mathbf{n}} = \hat{n}_{ij} \hat{n}_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$, where the components $\hat{n}_{ij} \hat{n}_{kl}$ can be expressed by:

$$\hat{n}_{ij} \hat{n}_{kl} = \begin{bmatrix} \hat{n}_{11} \hat{n}_{11} & \hat{n}_{11} \hat{n}_{22} & \hat{n}_{11} \hat{n}_{33} & \hat{n}_{11} \hat{n}_{12} & \hat{n}_{11} \hat{n}_{23} & \hat{n}_{11} \hat{n}_{13} \\ \hat{n}_{22} \hat{n}_{11} & \hat{n}_{22} \hat{n}_{22} & \hat{n}_{22} \hat{n}_{33} & \hat{n}_{22} \hat{n}_{12} & \hat{n}_{22} \hat{n}_{23} & \hat{n}_{22} \hat{n}_{13} \\ \hat{n}_{33} \hat{n}_{11} & \hat{n}_{33} \hat{n}_{22} & \hat{n}_{33} \hat{n}_{33} & \hat{n}_{33} \hat{n}_{12} & \hat{n}_{33} \hat{n}_{23} & \hat{n}_{33} \hat{n}_{13} \\ \hat{n}_{12} \hat{n}_{11} & \hat{n}_{12} \hat{n}_{22} & \hat{n}_{12} \hat{n}_{33} & \hat{n}_{12} \hat{n}_{12} & \hat{n}_{12} \hat{n}_{23} & \hat{n}_{12} \hat{n}_{13} \\ \hat{n}_{23} \hat{n}_{11} & \hat{n}_{23} \hat{n}_{22} & \hat{n}_{23} \hat{n}_{33} & \hat{n}_{23} \hat{n}_{12} & \hat{n}_{23} \hat{n}_{23} & \hat{n}_{23} \hat{n}_{13} \\ \hat{n}_{13} \hat{n}_{11} & \hat{n}_{13} \hat{n}_{22} & \hat{n}_{13} \hat{n}_{33} & \hat{n}_{13} \hat{n}_{12} & \hat{n}_{13} \hat{n}_{23} & \hat{n}_{13} \hat{n}_{13} \end{bmatrix} \quad (\text{C.16})$$

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